

LECTURE NOTES SK015

KOLEJ MATRIKULASI SARAWAK



CHAPTER 1

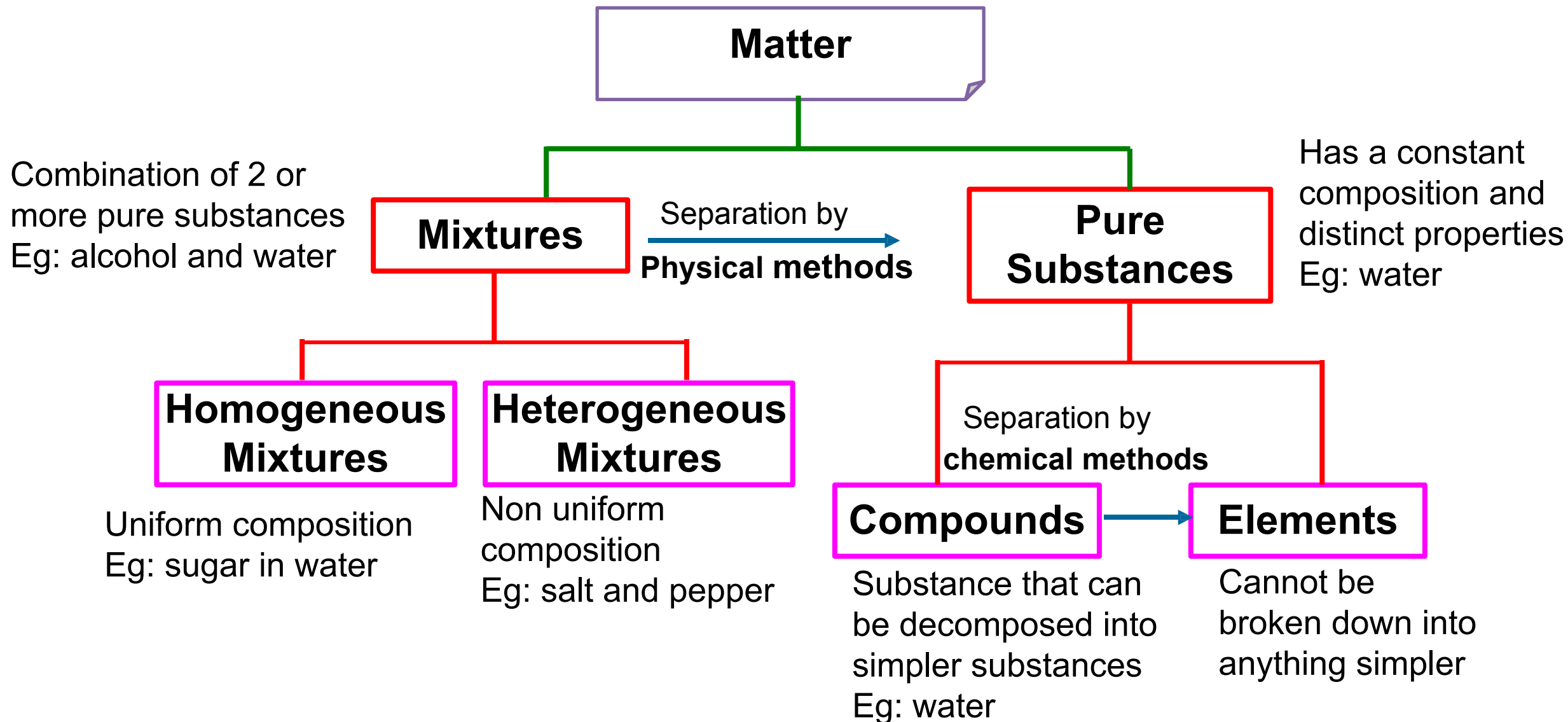
MATTER

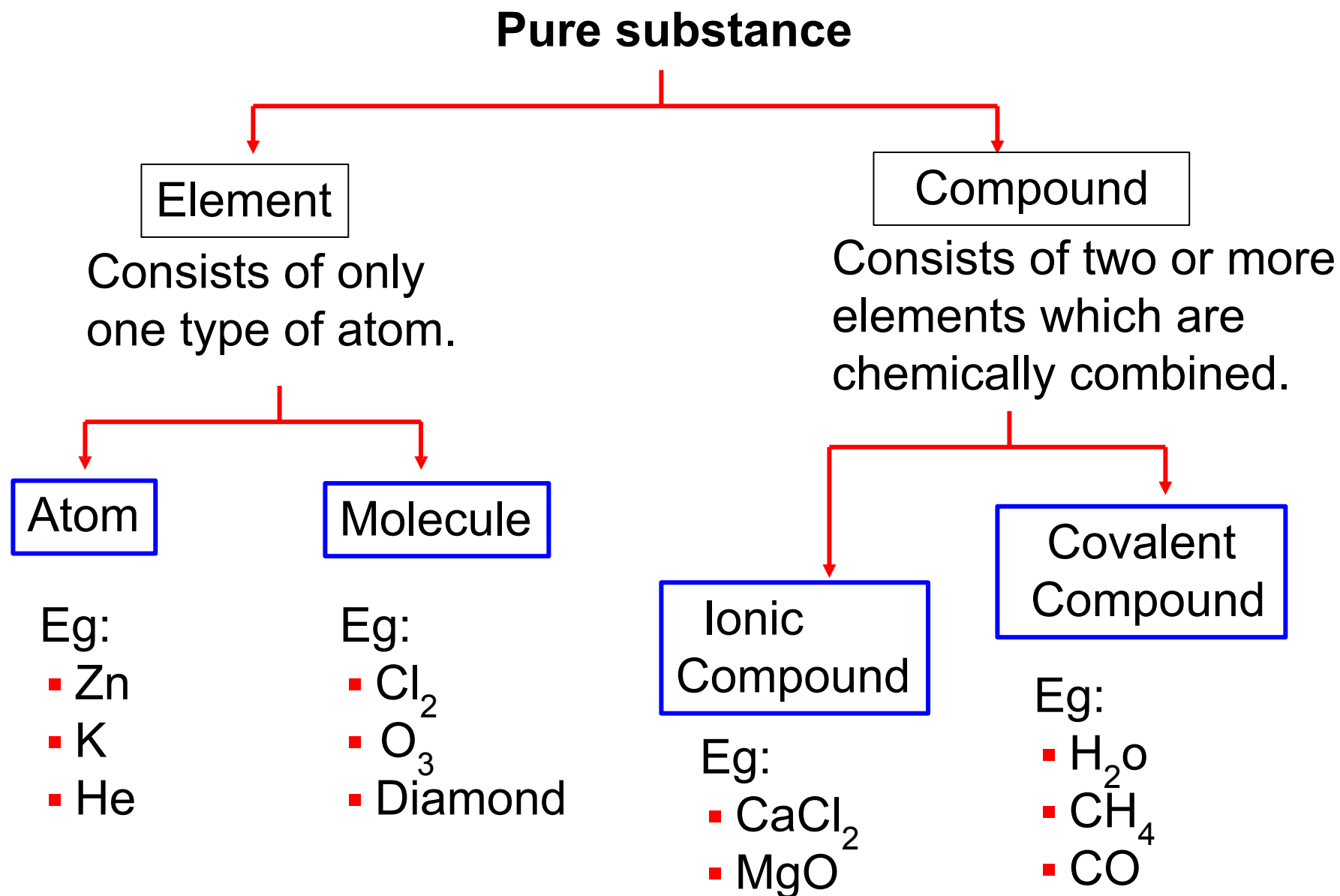
Chemistry is the study of matter and the changes that take place with that matter.

1.0 : Matter

- Made up **everything** on earth, **everything** in our solar system, **everything** in our galaxy, and **everything** in the universe.
- Name that scientists have given to everything that you can **touch**, or **see**, or **feel**, or **smell**.

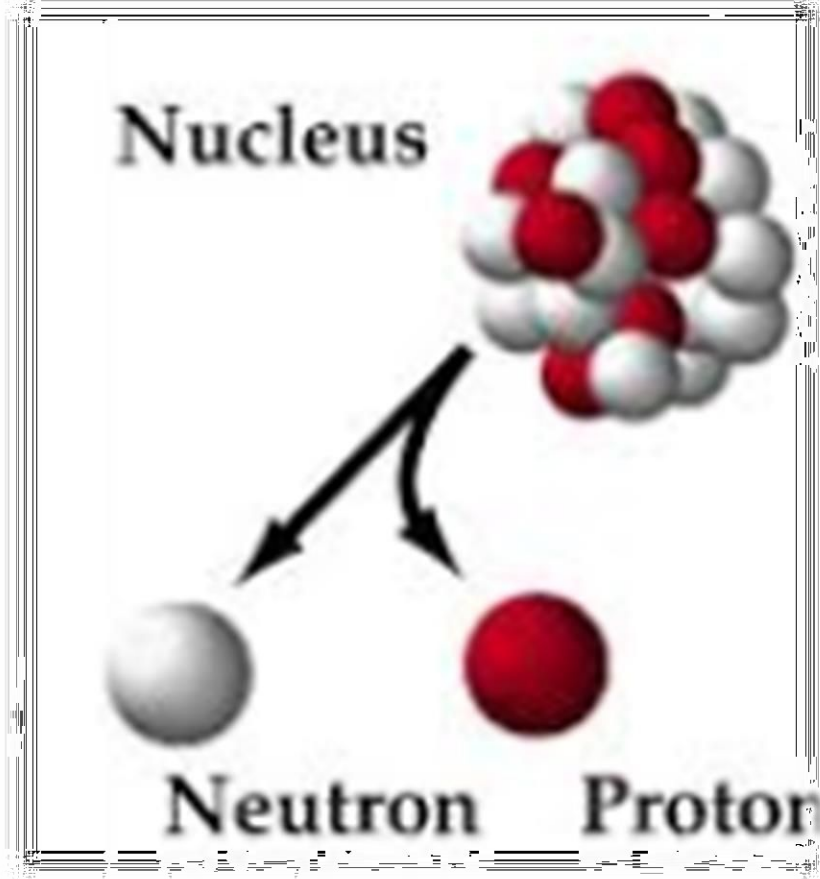
Classification of Matter





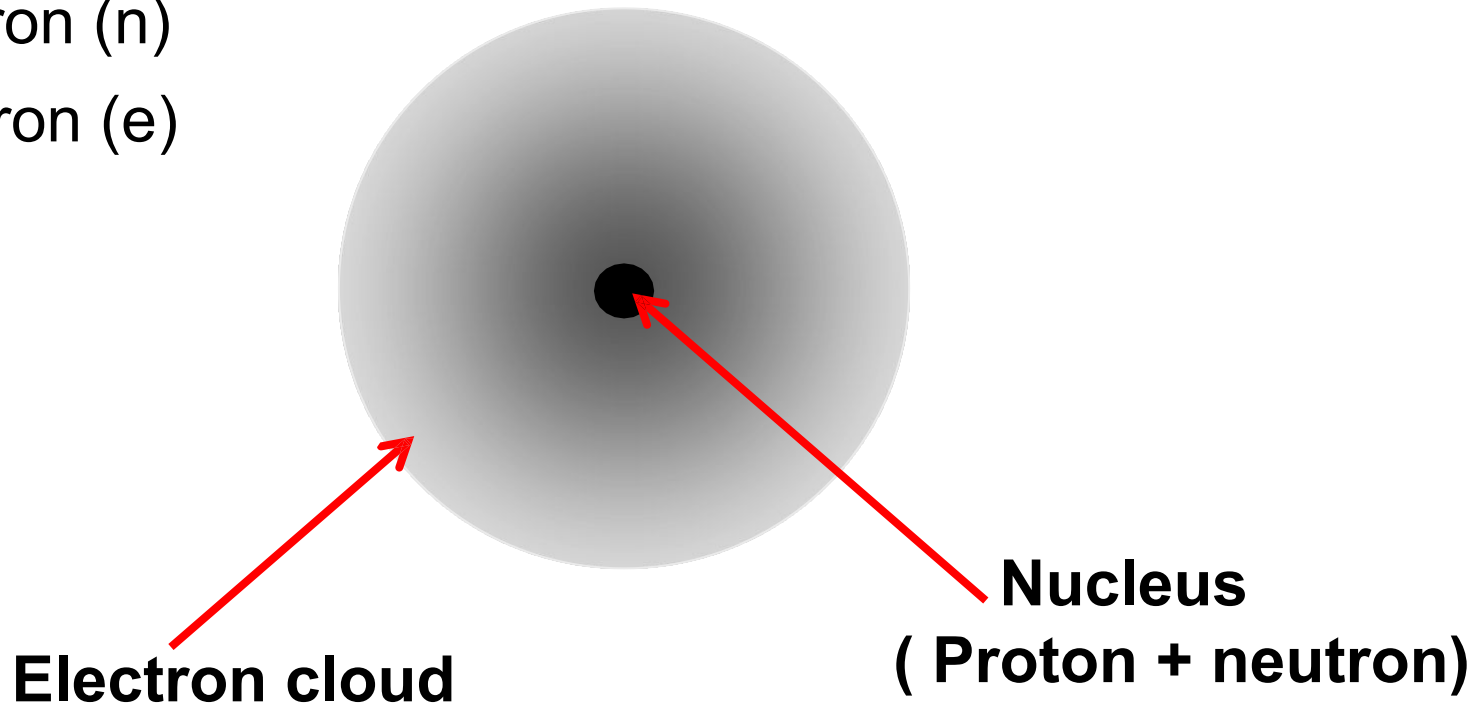
1.1

Atoms & Molecules



Model of an Atom

- ❑ An atom is the **smallest unit** of a chemical element/compound.
- ❑ In an atom, there are 3 subatomic particles:
 - Proton (p)
 - Neutron (n)
 - Electron (e)



- All neutral atoms can be identified by the number of protons they contain.

□ **Proton number, Z:**

- is the number of proton in the nucleus of an atom.
- also known as atomic number.

□ **Nucleon number, A:**

- is the total number of protons and neutrons found in the nucleus of the atom.
= proton number (Z) + number of neutrons.
- also known as mass number.



Atoms are neutral, therefore
No. of electrons = No. of protons

No of neutrons = $A - Z$

Isotopes

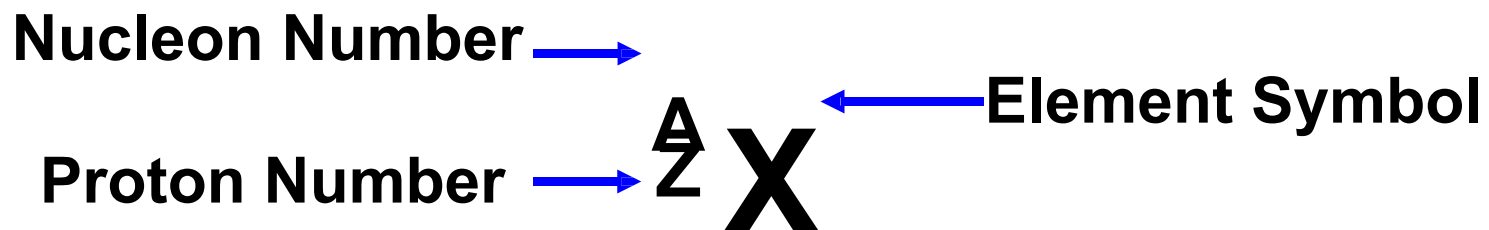
Isotopes are two or more atoms of the same element that have the **same number of protons** in their nucleus but **different number of neutrons**.

OR

Two or more atoms of the same element that have the **same proton number** but **different nucleon number**.

Isotopic Notation

The symbol for an atom:



Isotopes

**Isotopes of an element
have the same :**

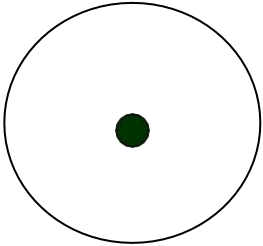
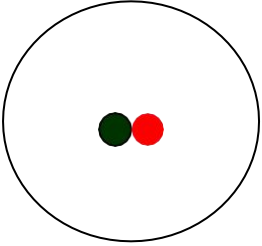
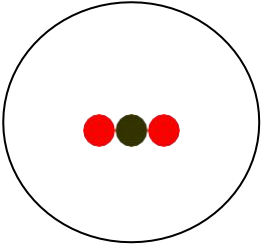
- ☐ **Proton number**
- ☐ **Number of electrons in a neutral atom**
- ☐ **Chemical properties**

**Isotopes of an element
have different :**

- ☐ **Nucleon number**
- ☐ **Number of neutron in an atom**
- ☐ **Physical properties such as densities**
- ☐ **Relative masses**

Example:

Hydrogen has three naturally occurring isotopes.

Structure of neutral atom			
Isotopic Notation	${}^1_1\text{H}$ protium	${}^2_1\text{H}$ deuterium	${}^3_1\text{H}$ tritium
Number of protons	1	1	1
Number of neutrons	0	1	2
Nucleon number (no. of protons + no. of neutrons)	1	2	3

Example:

Give the number of protons, neutrons and electrons in each of the following species:

Symbol	No. of Protons	No. of Neutrons	No. of Electrons
$^{235}_{92}\text{U}$	92	143	92
$^{200}_{80}\text{Hg}$	80	120	80
$^{17}_8\text{O}^{2-}$	8	9	10
$^{59}_{27}\text{Co}^{3+}$	27	32	24

Relative Atomic Mass (A_r)



- The actual mass of atom is very small (in range of 10^{-20} kg).
- Hence, relative atomic mass is used.

Relative atomic mass, A_r of an element :

mass of an atom in comparison $\frac{1}{12}$ of mass of carbon-12 atom
to

$$A_r \text{ of X} = \frac{\text{Average mass of one atom of X}}{\frac{1}{12} \times \text{mass of one atom of } ^{12}\text{C}}$$

- Relative atomic mass is a ratio, hence, it has **NO UNIT**.

Relative Molecular Mass (M_r)

- Mass of a molecule in comparison to 1/12 of mass of a carbon-12 atom.

$$M_r \text{ X} = \frac{\text{Average mass of one molecule of X}}{12} \times \text{mass of one atom of } ^{12}\text{C}$$

- M_r is sum of the relative atomic masses of all the atoms shown in the molecular formula.

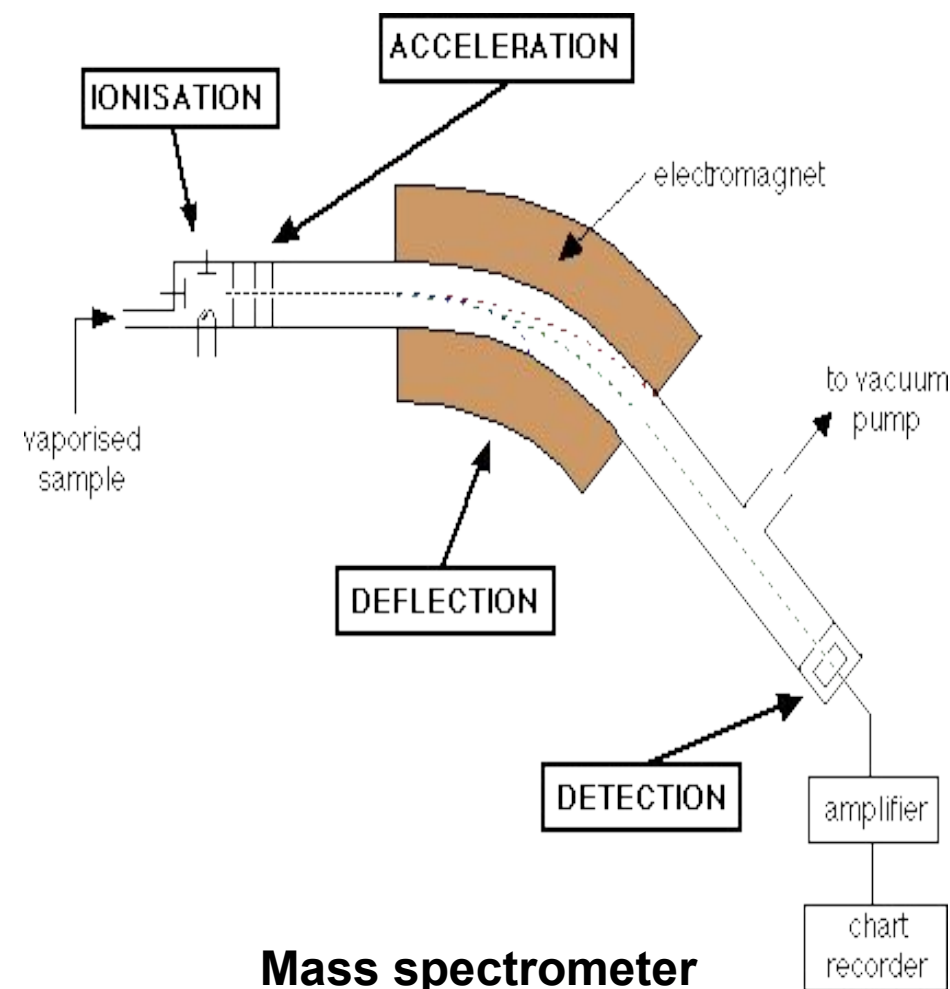
Measuring Atomic Mass

- An element occurs naturally as a mixture of isotopes.
- The atomic mass of an element is the average of its isotopic masses weighted according to their natural abundances and is determined by modern instruments such as mass spectrometer.

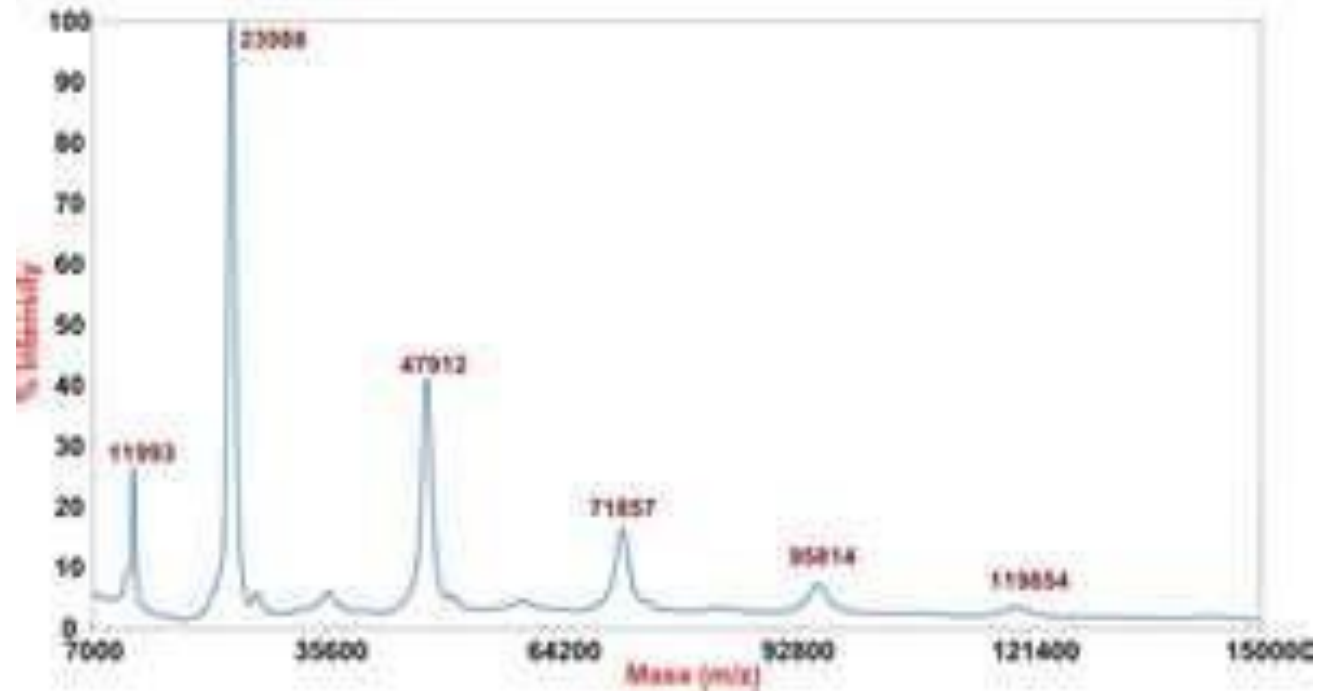
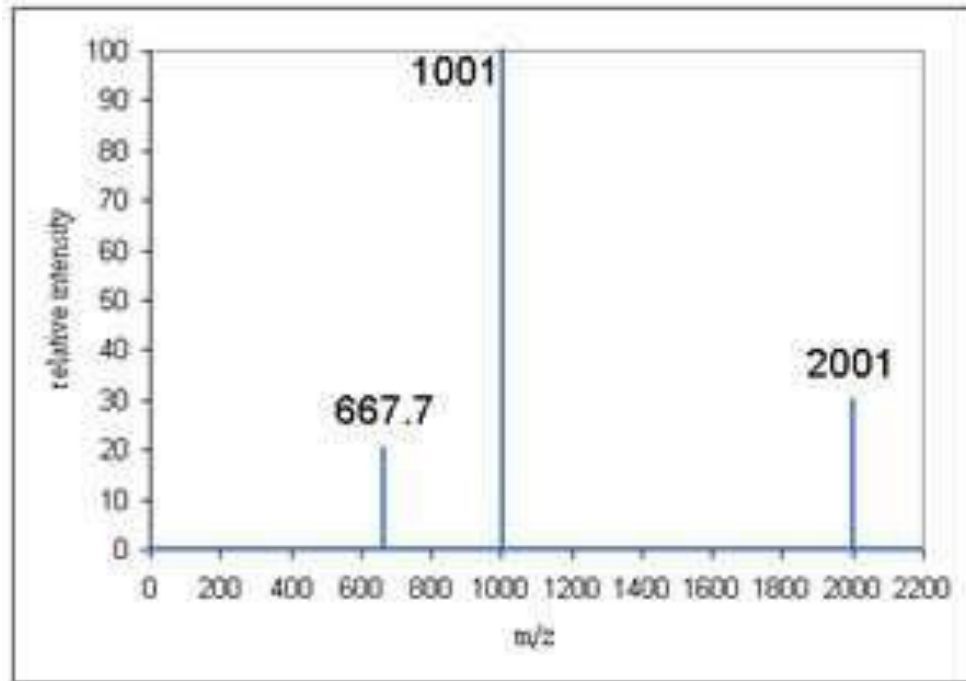
**Abundances = quantity or amount*

Mass spectrometer is used to determine:

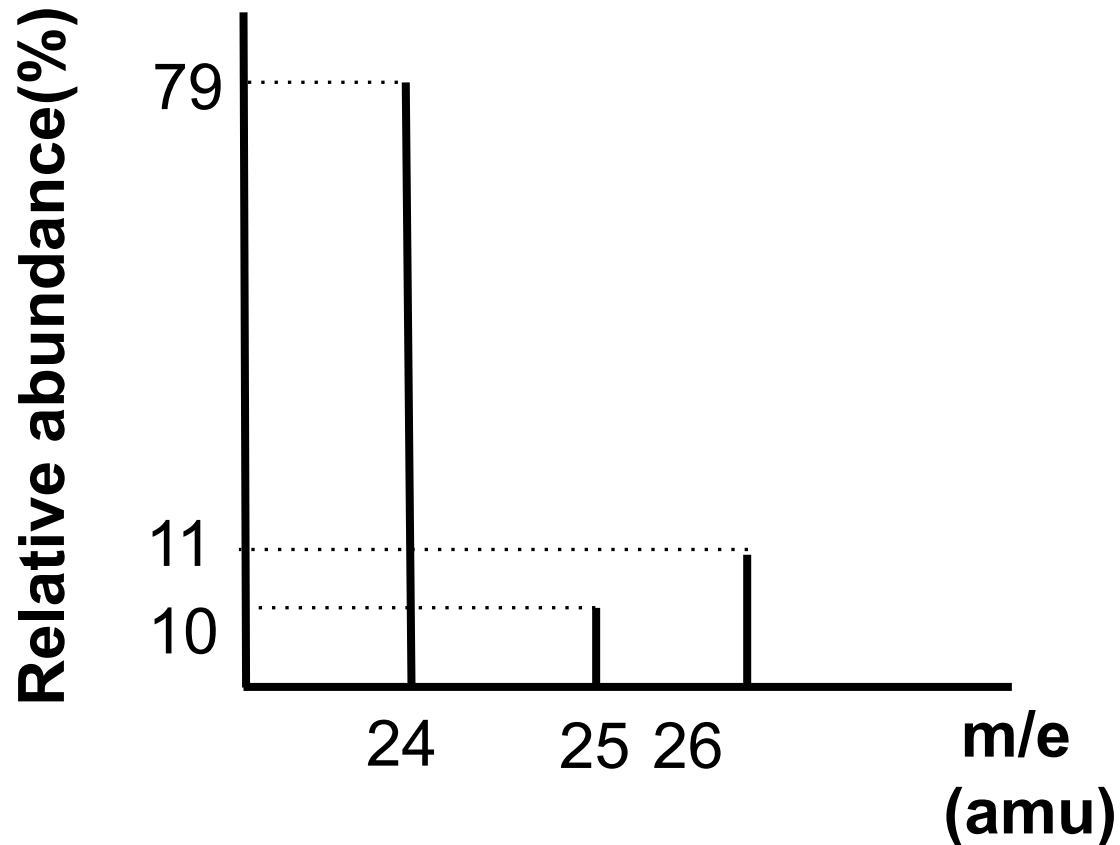
- Relative atomic mass (A_r) of an element.
- Relative molecular mass (M_r) of a compound.
- Types of isotopes that are found in the naturally occurring element including the abundance of the isotopes and its relative isotopic mass.
- The structure of the compound in an unknown sample.



A record of the separation of ions in a mass spectrometer is called a **mass spectrum**.



Example: Mass Spectrum of Magnesium



- The mass spectrum of Mg shows that Mg consists of **3 isotopes (3 peaks)**: ^{24}Mg , ^{25}Mg and ^{26}Mg .
- The **height of each line** is proportional to the **abundance of each isotope**.
- ^{24}Mg is the most abundant among the 3 isotopes

Interpretation of a mass spectrum:

Each peak represents a different isotope.

y-axis (relative abundance) shows the abundance of each isotope.

The relative height of each peak corresponds to the relative abundance of the isotopes

x-axis indicates the mass/charge ratio (m/e), which equals the mass divided by $+1$, for unipositive ions detected by spectrometer.

The m/e values are measured to identify the masses of different isotopes of an element

AVERAGE ATOMIC MASS OF AN ELEMENT

- Is the **average mass** of its **naturally occurring isotopes** weighted according to their **abundances**.

$$\text{Average atomic mass} = \frac{\sum Q_i m_i}{\sum Q_i}$$

Q_i = The relative abundance of an isotope of the element

m_i = The isotopic mass of the element

- Unit = amu @ u

Example 1: Mid Sem Session 2007/2008

Oxygen consists of three isotopes with atomic masses of 15.9949 amu, 16.9913 amu, and 17.9916 amu and their abundances are 99.76%, 0.0400% and 0.200% respectively.

Calculate the average atomic mass of oxygen.

[4 marks]

Solution :

$$\text{Average atomic mass of oxygen} = \frac{\sum Q_i m_i}{\sum Q_i}$$

$$= \frac{(99.76 \times 15.9949 \text{ amu}) + (0.0400 \times 16.9913 \text{ amu}) + (0.200 \times 17.9916 \text{ amu})}{99.76 + 0.0400 + 0.200}$$

$$= \mathbf{16.0 \text{ amu}}$$

Example 2: Mid Sem Session 2008/2009

The two naturally occurring isotopes of gallium are gallium-69 and gallium-71 which are found in the ratio of 3:2. Determine the atomic mass of gallium.

[3 marks]

Solution:

$$\begin{aligned}\text{Average atomic mass of gallium} &= \frac{\sum Q_i m_i}{\sum Q_i} \\ &= \frac{(3 \times 69 \text{ amu}) + (2 \times 71 \text{ amu})}{3 + 2} \\ &= \mathbf{69.8 \text{ amu}}\end{aligned}$$

Example 3: PSPM 2005/2006

The mass spectrum of copper is shown in **FIGURE 1**

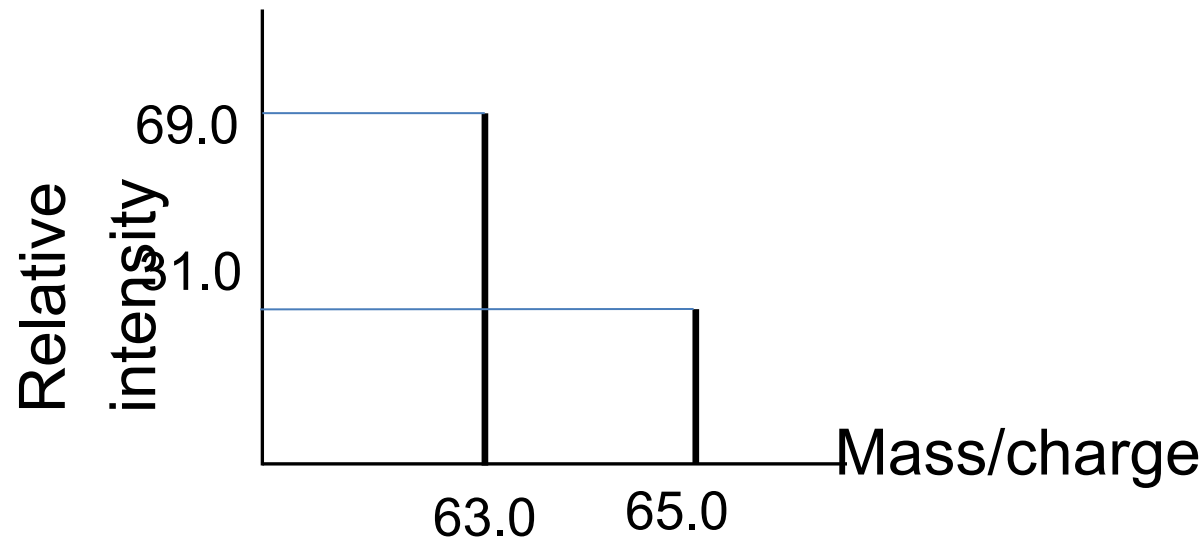


FIGURE 1

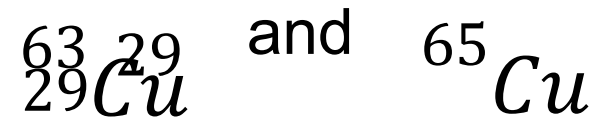
Based on **FIGURE 1**,

- Write all the isotopes of copper.
- Determine the percentage abundance of each isotope.
- Calculate the relative atomic mass of copper.

[4 marks]

Solution:

- i. Write the isotopic notation for all the isotopes of copper.



- ii. Determine the percentage abundance of each isotope.

$$\begin{aligned} \% {}^{63}\text{Cu} &= \frac{69}{69 + 31} \times 100\% \\ &= \mathbf{69\%} \end{aligned}$$

$$\begin{aligned} \% {}^{65}\text{Cu} &= \frac{31}{69 + 31} \times 100\% \\ &= \mathbf{31\%} \end{aligned}$$

Solution:

iii. Calculate the relative atomic mass of copper.

$$\begin{aligned}\text{Average atomic mass of Cu} &= \frac{\sum Q_i m_i}{\sum Q_i} \\ &= \frac{(69 \times 63 \text{ amu}) + (31 \times 65 \text{ amu})}{69 + 31} \\ &= 63.62 \text{ amu}\end{aligned}$$

$$\text{Relative atomic mass of Cu, } A_r \text{ Cu} = \frac{\text{Average mass of Cu}}{\frac{1}{12} \times \text{mass of one atom of } {}^{12}_6\text{C}}$$

$$= \frac{63.62 \text{ amu}}{\frac{1}{12} \times 12.00 \text{ amu}}$$

$$= \mathbf{63.62}$$

Example 4:

The atomic masses of ${}^6_3\text{Li}$ and ${}^7_3\text{Li}$ are 6.0151 amu and 7.0160 amu respectively. What is the relative abundance of each isotope if the relative atomic mass of lithium is 6.941?

Solution:

Let : Relative abundance of ${}^6\text{Li} = x$,
Relative abundance of ${}^7\text{Li} = 100\% - x$

$$\text{Average atomic mass} = \frac{\sum Q_i m_i}{\sum Q_i}$$

$$6.941 \text{ amu} = \frac{x(6.0151 \text{ amu}) + (100 - x)(7.0160 \text{ amu})}{100\%}$$

$$x = 7.49\%$$

$\therefore$ Relative abundance of ${}^6\text{Li} = \mathbf{7.49\%}$

$\therefore$ Relative abundance of ${}^7\text{Li} = 100 - 7.49 = \mathbf{92.5\%}$

Alternative method

Let : Relative abundance of ${}^6\text{Li} = x$

Relative abundance of ${}^7\text{Li} = 1 - x$

$$\text{Average atomic mass} = \frac{\sum Q_i m_i}{\sum Q_i}$$

$$6.941 \text{ amu} = \frac{x(6.0151 \text{ amu}) + (1 - x)(7.0160 \text{ amu})}{1}$$

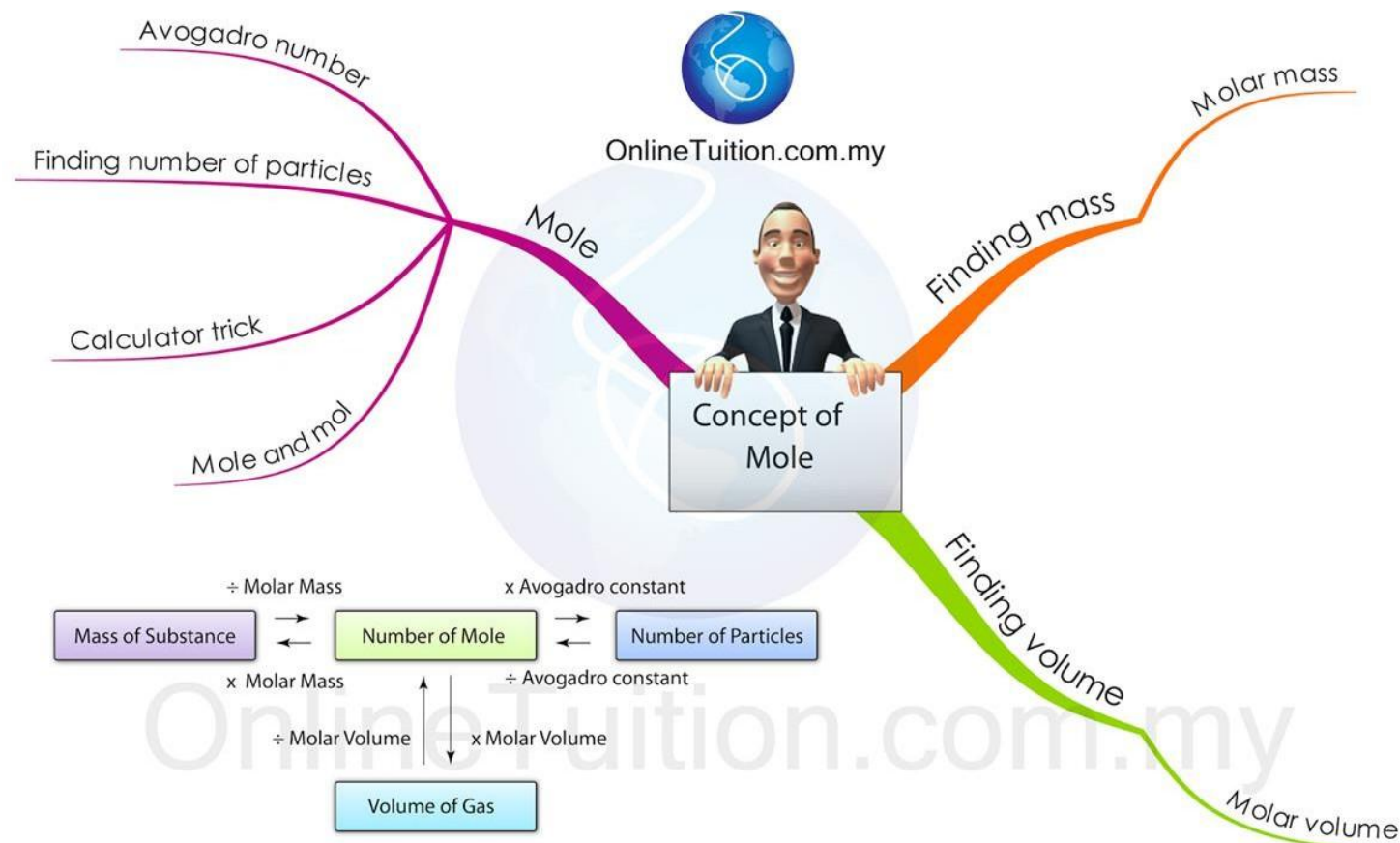
$$x = 0.0749$$

$\therefore$ Relative abundance of ${}^6\text{Li} = \mathbf{0.0749}$

$\therefore$ Relative abundance of ${}^7\text{Li} = 1 - 0.0749 = \mathbf{0.9251}$

1.2

Mole Concept



PRIOR KNOWLEDGE

One mole is the **amount of a substance** that contains the **same number of particles** that exists in exactly **12.00 g of carbon-12**.



1 mol of carbon-12 has a mass of 12.00 g and contains 6.02×10^{23} atoms.



Avogadro's number, $N_A = 6.02 \times 10^{23}$



Molar mass is the mass of a substance (in grams) that contains 6.02×10^{23} particles and has units of g mol^{-1} .

PRIOR KNOWLEDGE

- For any element, **molar mass** has the same value as **atomic mass** or **relative atomic mass**.

Example

Atomic mass of ^{12}C = 12.00 amu

Relative atomic mass of ^{12}C = 12.00

Molar mass of ^{12}C atoms = 12.00 g mol⁻¹

- Relationship between mole, mass, A_r , M_r and amount of particles.

Example

Molecular mass of SO_2 = 64 amu

Relative molecular mass, M_r of SO_2 = $32 + 2(16) = 64$

Molar mass of SO_2 = 64 g mol⁻¹

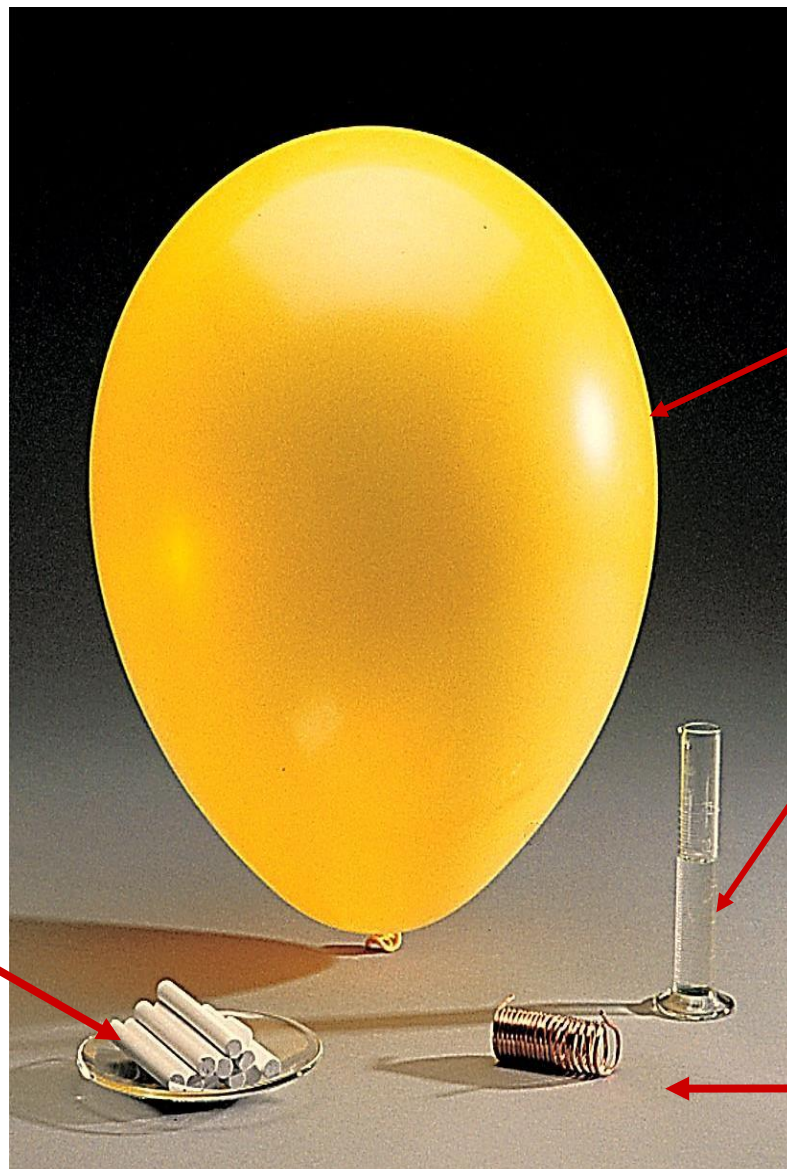
Mass of 1.0 mol of SO_2 or **6.02×10^{23} SO_2 molecules** = 64 g

Counting Atoms by Weighing Them

1 Mole of
 CaCO_3

6.02×10^{23} formula
units of CaCO_3
= 100.10 g

$\therefore$ Molar mass of CaCO_3
= 100.10 g mol⁻¹



1 mole oxygen

6.02×10^{23} of
 O_2 molecules
= 32.00 g
 O_2

$\therefore$ Molar mass of O_2
= 32.00 g mol⁻¹

1 mole of Water

6.02×10^{23} of
 H_2O molecules
= 18.00 g

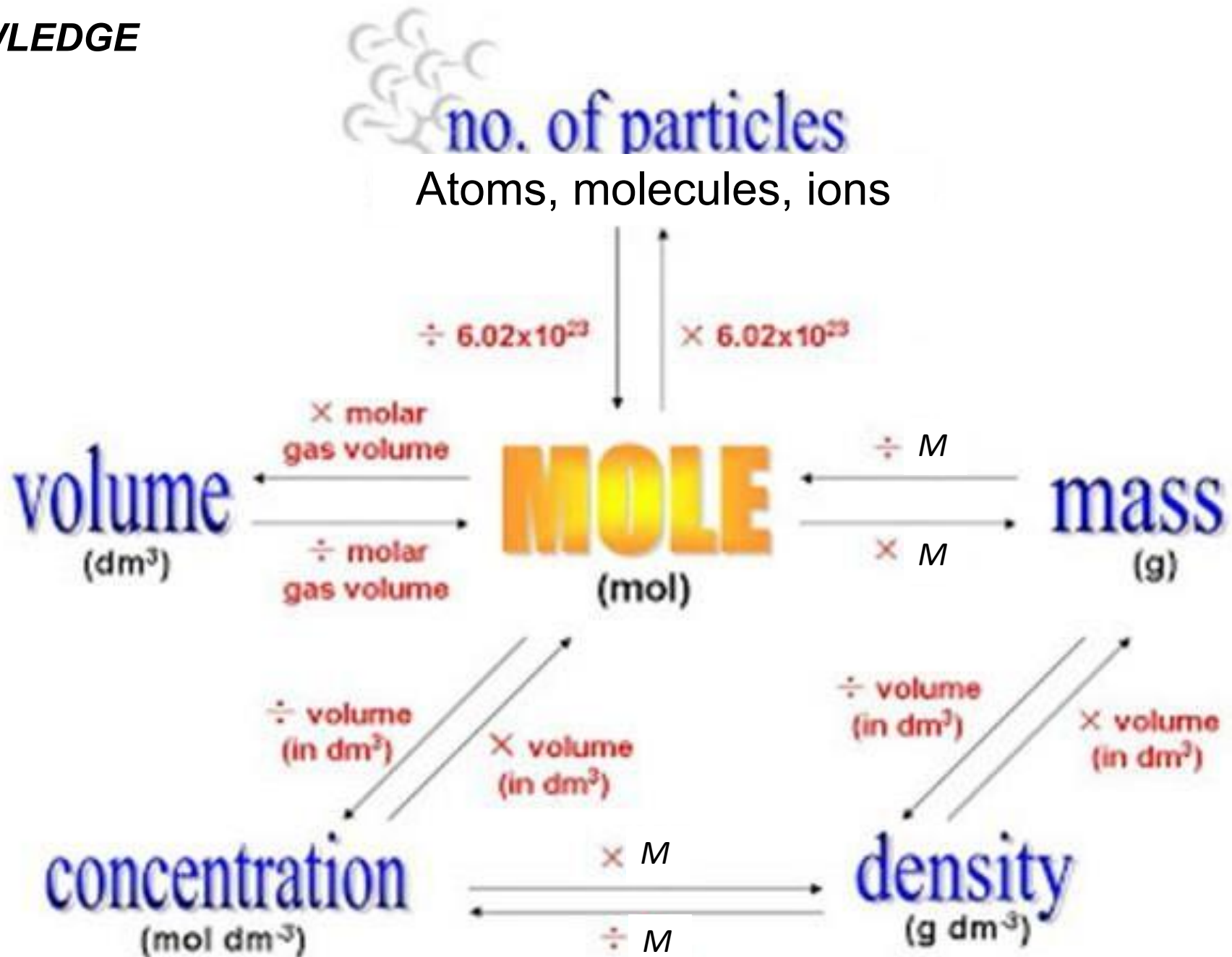
$\therefore$ Molar mass of H_2O
= 18.00 g mol⁻¹

1 mol of Copper

6.02×10^{23} of Cu atom
= 63.60
g

$\therefore$ Molar mass of Cu
= 63.60 g mol⁻¹

PRIOR KNOWLEDGE



PRIOR KNOWLEDGE

Mole Concept

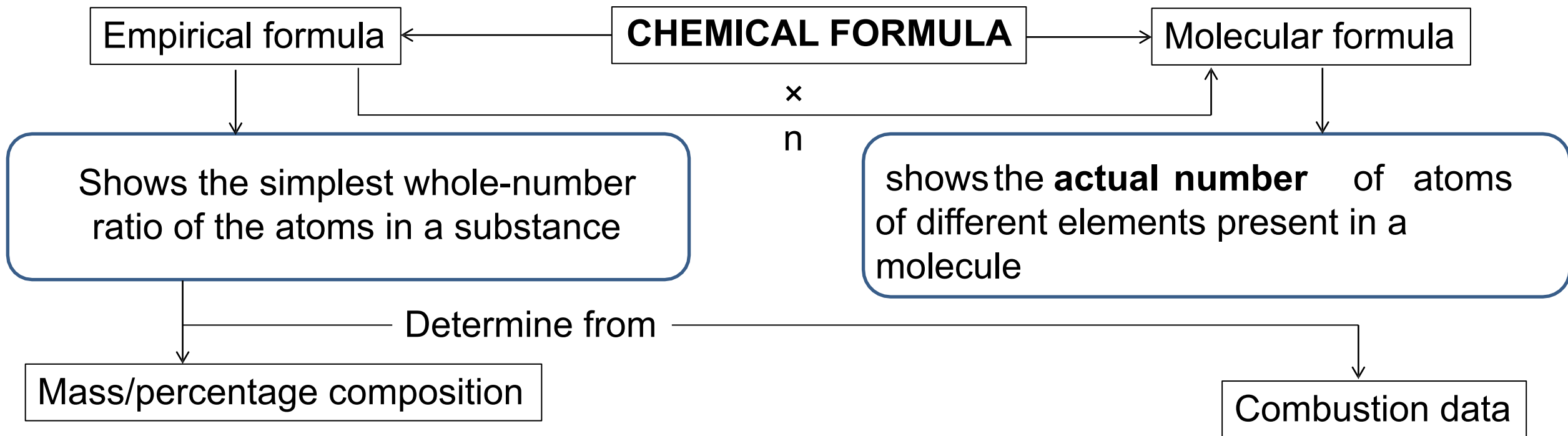
$$\text{Moles} = \frac{\text{Number of particles}}{N_A}$$

$$\text{Moles} = \frac{\text{Mass}}{\text{Molar mass}} \Rightarrow n = \frac{m}{M}$$

$$\text{Moles} = \frac{\text{Gas volume}}{\text{Molar volume}} \Rightarrow n = \frac{V}{V_m}$$

Molar volume of gases, $V_m = 22.4 \text{ dm}^3 \text{ mol}^{-1}$ at STP

$= 24 \text{ dm}^3 \text{ mol}^{-1}$ at room temperature



1. Assume a definite starting quantity (usually 100g) of the compound, if percentage composition is given.
2. Draw a table to tabulate the data.
3. Convert gram of element into mole.
4. Divide each mole in step 3 by the smallest of these values to get the simplest ratio.
5. Multiple the value obtained in step 4 by a number that will convert them to whole number.
6. Use these whole numbers as subscripts in the empirical formula.

❖ Use for organic compounds which contain carbon, hydrogen and oxygen or other elements.

Example 1:

Determine the empirical formula of a compound contains 2.1% H, 65.3% O and 32.6% S

Solution:

- Assume mass of compound is 100g

Element	H	O	S
Mass(g)	2.1	65.3	32.6
Number of moles (mol)	$\frac{2.1}{1} = 2.1$	$\frac{65.3}{16} = 4.0813$	$\frac{32.6}{32.1} = 1.0156$
Simplest ratio	$\frac{2.1}{1.0156} = 2$	$\frac{4.0813}{1.0156} = 4.0186$ ≈ 4	$\frac{1.0156}{1.0156} = 1$

**Empirical
formula:
 H_2SO_4**

Example 2:

1.08 g of aluminium combines chemically with 0.96 g of oxygen to form an oxide.
What is the empirical formula of the oxide? [Relative atomic mass: O = 16; Al = 27]

Solution:

Element	Al	O
Mass(g)	1.08	0.96
Number of moles (mol)	$\frac{1.08}{27} = 0.04$	$\frac{0.96}{16} = 0.06$
Simplest ratio	$\frac{0.04}{0.04} = 1$	$\frac{0.06}{0.04} = 1.5$
	$1 \times 2 = 2$	$1.5 \times 2 = 3$

Empirical formula: Al_2O_3

Example 3:

The composition of an organic compound is 38.71% C, 9.68% H and 51.61% O. Its relative molecular mass is 62. What are the empirical formula and molecular formula of the compound?

Solution:

- Assume mass of compound is 100g

Empirical formula:
CH₃O

Molecular
formula: (CH₃O)_n

$$\begin{aligned}(\text{Empirical formula})n &= M_r \\ (\text{CH}_3\text{O})n &= 62 \\ (12+3+16)n &= 62 \\ n &= 2\end{aligned}$$

Molecular formula:
C₂H₆O₂

Element	C	H	O
Mass(g)	38.71	9.68	51.61
Number of moles (mol)	$\frac{38.71}{12} = 3.2258$	$\frac{9.68}{1} = 9.68$	$\frac{51.61}{16} = 3.2256$
Simplest ratio	$\frac{3.2256}{3.2256} = 1$	$\frac{9.68}{3.2256} = 3$	$\frac{3.2256}{3.2256} = 1$

Example 4:

3.26 g of Ti_xCl_y was reduced by Na. If 0.81g of Ti was obtained, determine the value of x and y in the compound Ti_xCl_y . (Ar: Ti = 48 ; Cl = 35.5)

Solution:

Mass of Cl = $3.26 - 0.81 = 2.45$ g

Element	Ti	Cl
Mass(g)	0.81	2.45
Number of moles (mol)	$\frac{0.81}{48} = 1.69 \times 10^{-2}$	$\frac{2.45}{35.5} = 6.90 \times 10^{-2}$
Simplest ratio	$\frac{1.69 \times 10^{-2}}{1.69 \times 10^{-2}} = 1$	$\frac{6.90 \times 10^{-2}}{1.69 \times 10^{-2}} = 4.08 \approx 4$

$\therefore x = 1, y = 4$

Example 5:

Combustion of 2.30 g of an organic sample, X, yields 3.30 g CO₂ and 1.80 g of H₂O. Determine the empirical formula of X.

Solution:

Tips: mass of X containing C, H and O = 2.30 g

Find out the mass composition of C, H and O respectively in the sample!

Mass of C:

Molar mass of CO₂ = 12 + (2 x 16) = 44 g/mol

1 mol CO₂ ≡ 1 mol C

44 g CO₂ ≡ 12 g C

$$\begin{array}{l} \therefore 3.30 \text{ g CO}_2 \\ \equiv \end{array} \quad \frac{12 \text{ g C} \times 3.30 \text{ g CO}_2}{44 \text{ g CO}_2} = 0.90 \text{ g C}$$

Mass of H:

$$\text{Molar mass of H}_2\text{O} = (2 \times 1) + 16 = 18 \text{ g/mol}$$

$$1 \text{ mol H}_2\text{O} \equiv 2 \text{ mol H}$$

$$\therefore \frac{18 \text{ g H}_2\text{O}}{1.80 \text{ g H}_2\text{O}} \equiv \frac{2 \text{ g H}}{18 \text{ g H}_2\text{O}} \times 1.80 \text{ g H}_2\text{O} = 0.2 \text{ g H}$$

Mass of O:

$$\text{Mass of C} + \text{mass of H} + \text{mass of O} = 2.30 \text{ g}$$

$$\begin{aligned} \text{Mass of O} &= 2.30 - 0.9 - 0.2 \\ &= 1.20 \text{ g O} \end{aligned}$$

Element	C	H	O
Mass (g)	0.90	0.20	1.20
Number of moles (mol)	$\frac{0.90}{12} = 0.075$	$\frac{0.20}{1} = 0.2$	$\frac{1.20}{16} = 0.075$
Simplest mole ratio	$\frac{0.075}{0.075} = 1$	$\frac{0.20}{0.075} = 2.667$	$\frac{0.075}{0.075} = 1$
	$1 \times 3 = 3$	$2.667 \times 3 = 8$	$1 \times 3 = 3$

Therefore, the empirical formula is = $\text{C}_3\text{H}_8\text{O}_3$

Keep in mind !

Never round off values close to whole number in order to get a simplest ratio, but ***multiply*** the value by a factor until you get a whole number.

Concentration of a solution

is the amount of solute present in a given quantity of solvent in a solution.

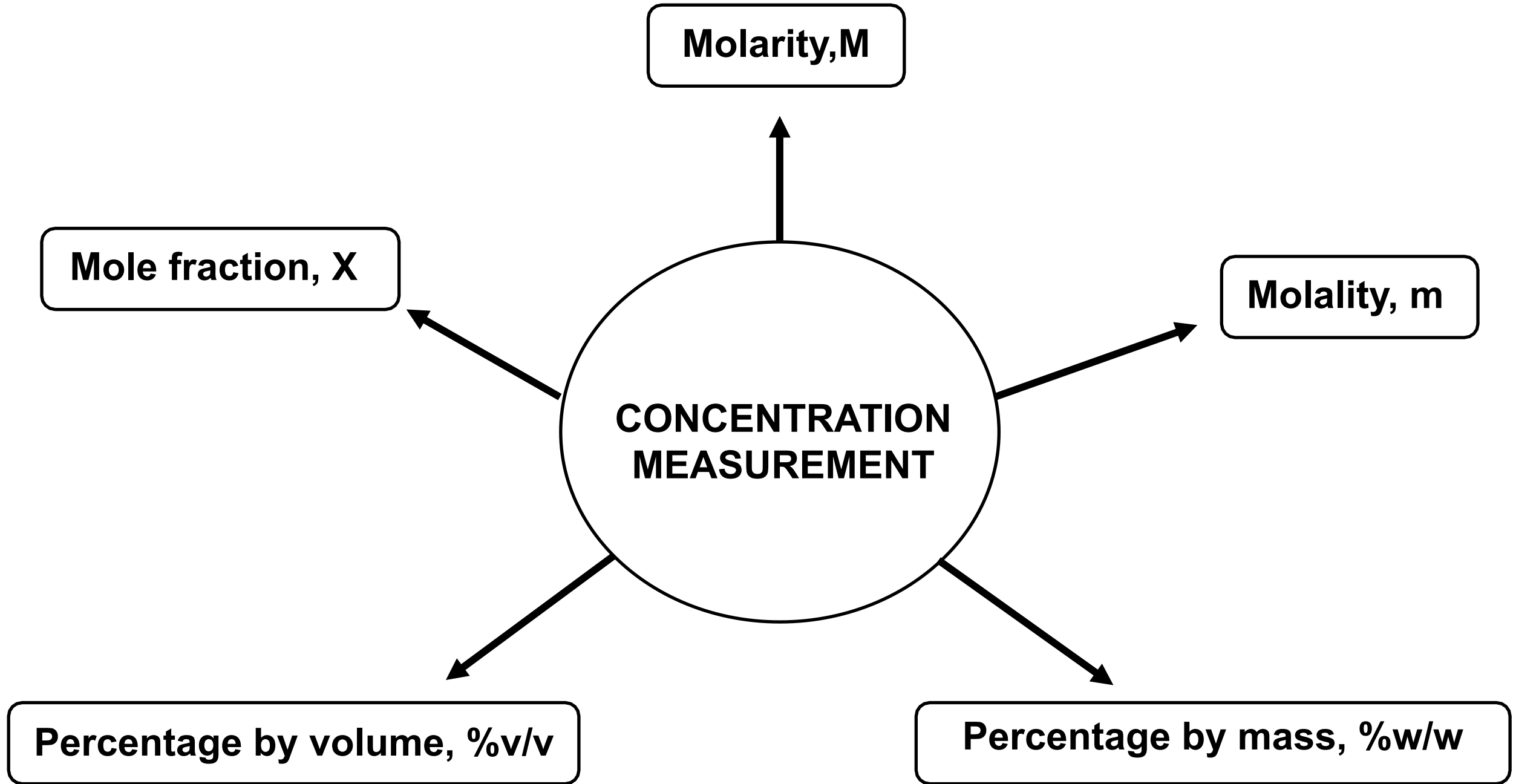
Solution is a **homogenous mixture** formed when an amount of solute dissolves completely in a solvent.

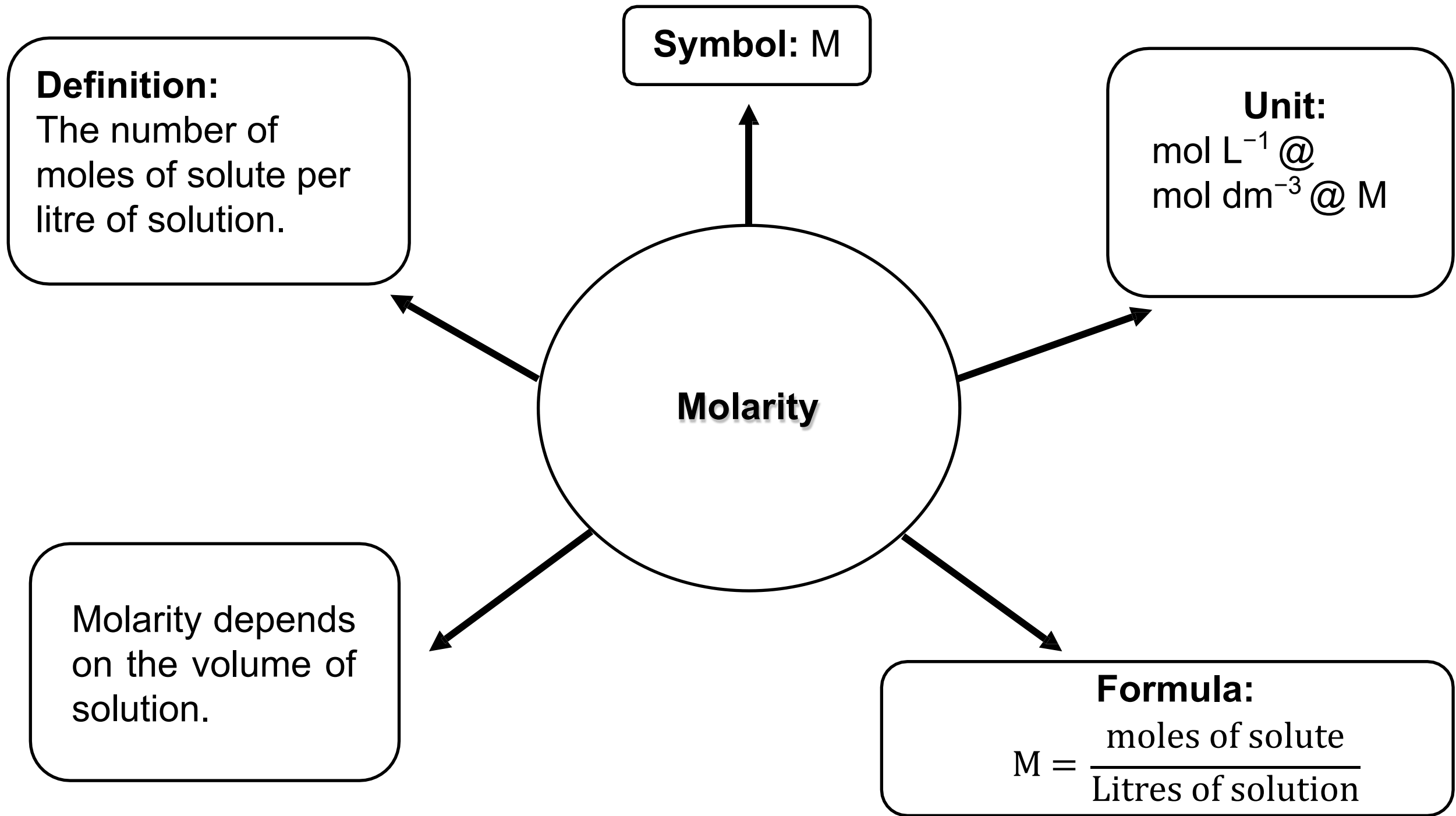
Solute is the substance being dissolved and present in the **smaller amount** in a solution.

Solvent is the substance doing the solving and present in the **larger amount** in a solution.

Example:







Definition:

The number of mole of solute per kilograms of solvent.

Symbol: m

Unit:

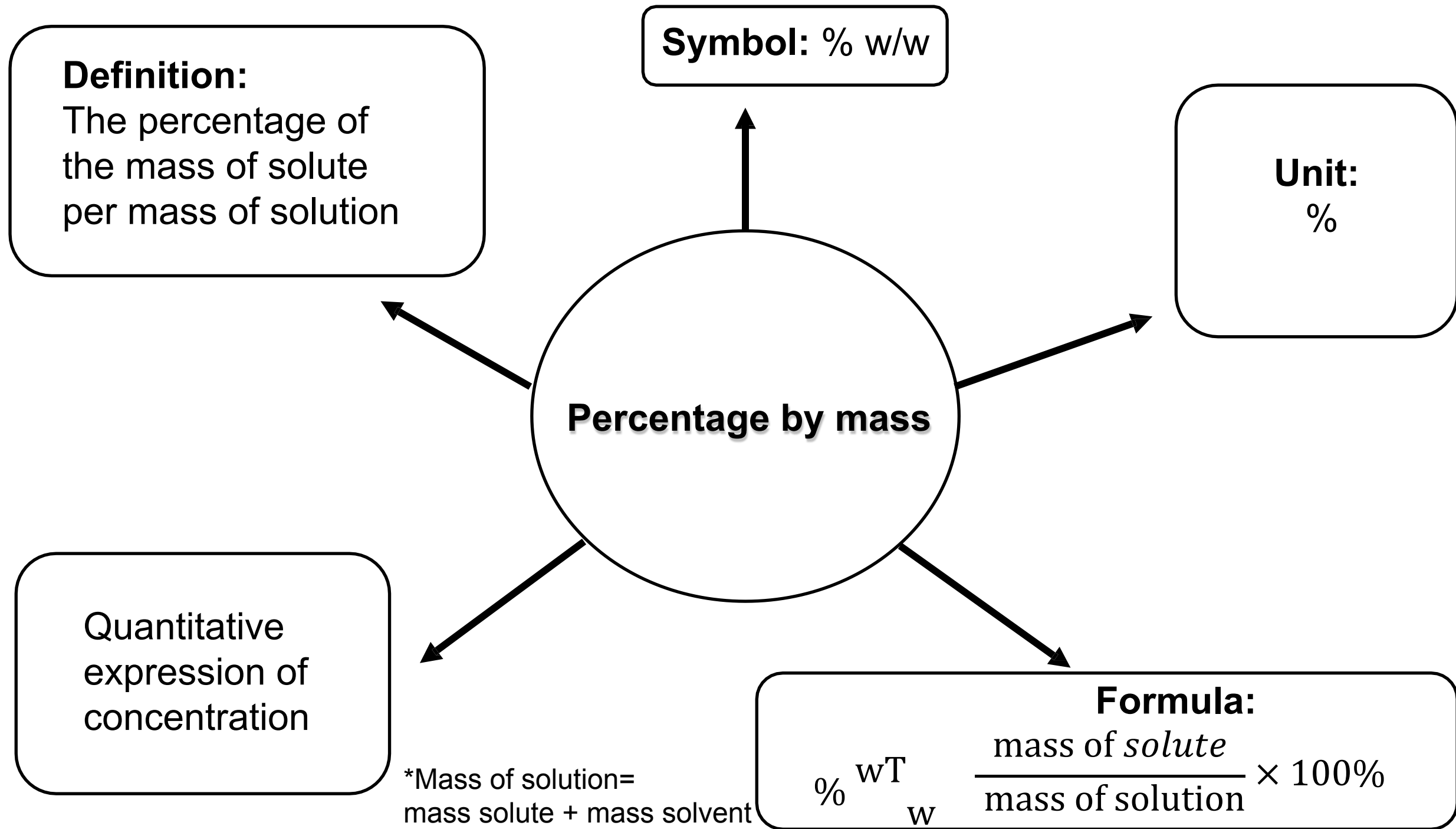
mol kg^{-1} @
molal

Molality

Molality depends on mass of solvent.

Formula:

$$m = \frac{\text{Moles of solute}}{\text{Kilograms of solvent}}$$



Symbol: % v/v

Unit:
%

**Percentage by
volume**

Definition:

The percentage of the volume of solute per volume of solution

Often used to describe the concentration of a solution that was prepared by dissolving a liquid in some solvent.

Formula:

$$\% \text{ v/v} = \frac{\text{volume of solute}}{\text{volume of solution}} \times 100\%$$

Symbol: X

Unit:
no unit

Mole Fraction

Formula:

$$X_A = \frac{n_A}{n_{\text{Total}}}$$

Definition:

Mole fraction is the ratio of the number of moles of the species to the total of number of moles present.

Mole fraction is a useful quantity for analysing gas mixtures in conjunction with Dalton's law of partial pressures

Example 1:

A solution is made by dissolving 2.355 g of sulfuric acid, H_2SO_4 in water. The total volume of the solution is 50.0 mL. What is the molar concentration of sulfuric acid?

Solution:

$$\frac{\text{mol H}_2\text{SO}_4}{98.1} = \frac{2.355}{98.1}$$

$$\text{Molarity of H}_2\text{SO}_4 = \frac{2.401 \times 10^{-2} \text{ moles of H}_2\text{SO}_4}{50 \text{ mL} \times \frac{1 \text{ L}}{1000 \text{ mL}}} = \frac{2.401 \times 10^{-2} \text{ mol}}{\text{Litres of solution}}$$

$$= \frac{2.401 \times 10^{-2} \text{ mol}}{0.0500 \text{ L}}$$

$$= \mathbf{0.4802 \text{ M}}$$

$$50 \text{ mL} = 0.0500 \text{ L}$$

Example 2 :

Copper sulphate, CuSO_4 is a bluish inorganic compound. It is soluble in water. Copper sulphate is used as a chemical reagent in Fehling's solution and Benedict's solution. In a CuSO_4 solution, CuSO_4 and water content is 100 g and 900 g respectively. If the molar mass of CuSO_4 is 159.7 g mol^{-1} , Calculate

- a) the molal concentration of CuSO_4 .
- b) the mole fraction of each component.
- c) the percentage by mass of CuSO_4 .

Solution:

$$\text{a) Molality of } \text{CuSO}_4 = \frac{\text{moles of } \text{CuSO}_4}{\text{kilograms of solvent}}$$

$$\text{Moles of } \text{CuSO}_4 = \frac{100}{159.7} = 0.626$$

mol

$$900 \text{ g} = 0.9 \text{ kg}$$

$$\therefore \text{Molality of } \text{CuSO}_4 = \frac{0.626 \text{ mol}}{0.9 \text{ kg}}$$

$$= 0.696$$

molal

b) The mole fraction of each component.

$$X_{\text{CuSO}_4} = \frac{n_{\text{CuSO}_4}}{n_{\text{CuSO}_4} + n_{\text{H}_2\text{O}}}$$

$$n_{\text{H}_2\text{O}} = \frac{900\text{g}}{18\text{g mol}^{-1}} = 50\text{ mol}$$

$$n_{\text{CuSO}_4} = 0.626\text{ mol}$$

$$X_{\text{CuSO}_4} = \frac{0.626}{0.626 + 50}$$

$$= \mathbf{0.012}$$

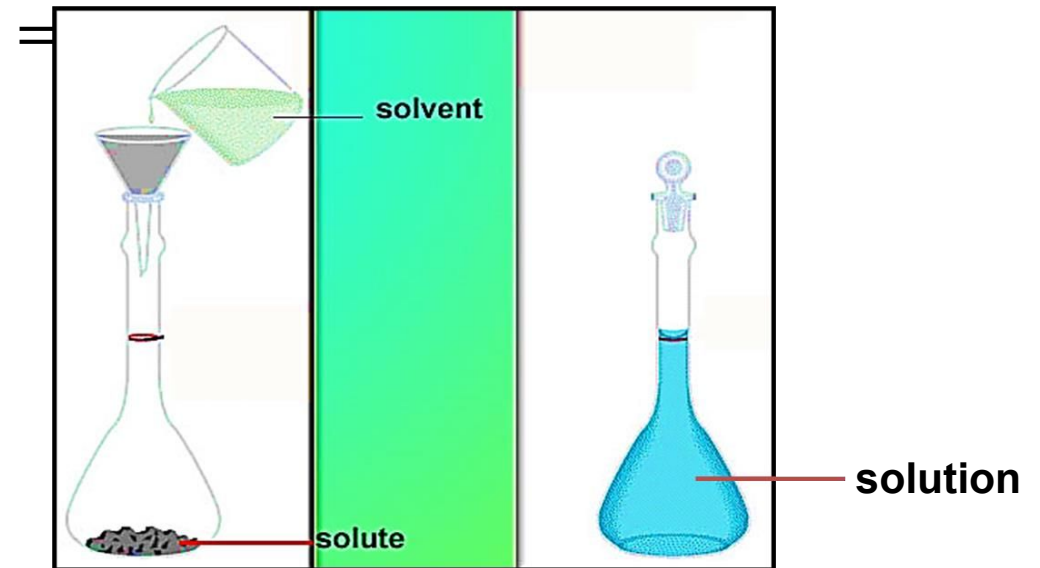
$$X_{\text{H}_2\text{O}} = 1 -$$

$$X_{\text{CuSO}_4} = 1 - 0.012$$

c) The percentage by mass of CuSO_4 .

$$\% \text{ wT}_w \text{ of CuSO}_4 = \frac{\text{mass of CuSO}_4}{\text{mass of solution}} \times 100\%$$

$$= \frac{100}{900} \times 100\% = \mathbf{10\%}$$



Example 3:

Since exact volume or mass is not given.

Concentrated hydrochloric acid solution is 36.5 % HCl by mass. Its density is 1.18 g mL^{-1} . Calculate

- a) the molality and
- b) the molarity of the solution

Solution:

a) 36.5 % HCl $\Rightarrow$ **Assume 36.5 g HCl in 100 g**

solution

Then,

$$\text{mass of solution} = 100 \text{ g}$$

$$\text{mass of HCl} = \frac{36.5}{100} \times 100 = 36.5 \text{ g}$$

$$\text{Molar mass of HCl} = 35.5 + 1.0 = 36.5 \text{ g mol}^{-1}$$

$$\text{mass of H}_2\text{O} = 100 - 36.5 = 63.5 \text{ g (solvent)}$$

$$n_{\text{HCl}} = \frac{36.5 \text{ g}}{36.5} = 1 \text{ mol}$$

Continue:

$$\begin{aligned} \text{(a) Molality} &= \frac{\text{moles of HCl}}{\text{kilograms of solvent}} \\ &= \frac{1 \text{ mol}}{63.5 \times 10^{-3} \text{ kg}} = \mathbf{15.7 \text{ mol kg}^{-1}} \end{aligned}$$

Solution:

b) Molarity
=

moles HCl
litres of solution


$$\rho_{\text{solution}} = \frac{m_{\text{solution}}}{V_{\text{solution}}}$$

$$1.18 \text{ g mL}^{-1} = \frac{100 \text{ g}}{V_{\text{solution}}}$$

$$\begin{aligned} V_{\text{solution}} &= \frac{100 \text{ g}}{1.18 \text{ g mL}^{-1}} \\ &= 84.75 \text{ mL} \end{aligned}$$

$$\begin{aligned} \text{Molarity} &= \frac{1 \text{ mol}}{84.75 \times 10^{-3} \text{ L}} \\ &= \end{aligned}$$

$$= 11.8 \text{ mol L}^{-1}$$

Example 4:

Rubbing alcohol is commonly used as an antiseptic for small cuts. It is sold as 70% (v/v) solution of isopropyl alcohol in water. What volume of isopropyl alcohol is used to make 500 mL of rubbing alcohol?

Solution:

$$\% \text{ v/v} = \frac{\text{Volume of solute}}{\text{Volume of solution}} \times 100\%$$

Solute: isopropyl alcohol

$$\% \text{ v/v} = \frac{\text{Volume of isopropyl alcohol}}{\text{Volume of solution}} \times 100\%$$

$$70 = \frac{\text{Volume of isopropyl alcohol}}{500} \times 100\%$$

$\therefore$ Volume of isopropyl alcohol = **350 mL**



Inspection
method

Chemical
equation

Ion-electron
method

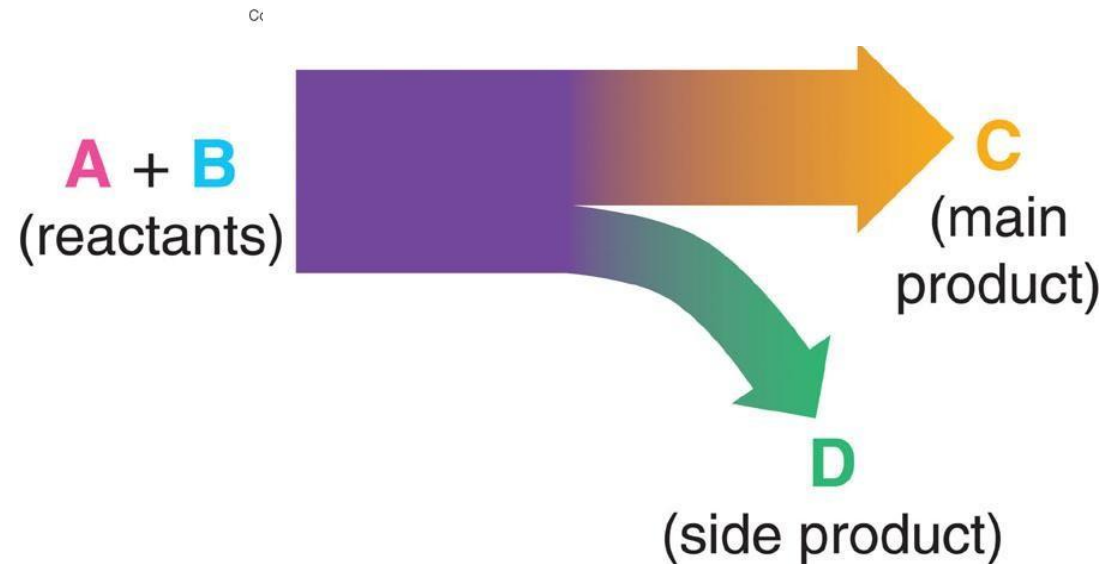
1.3 Stoichiometry

Limiting reactant

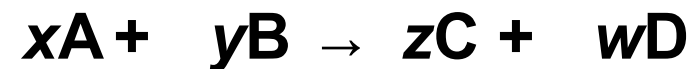
Calculation

Percentage
yield

Balancing Chemical Equation

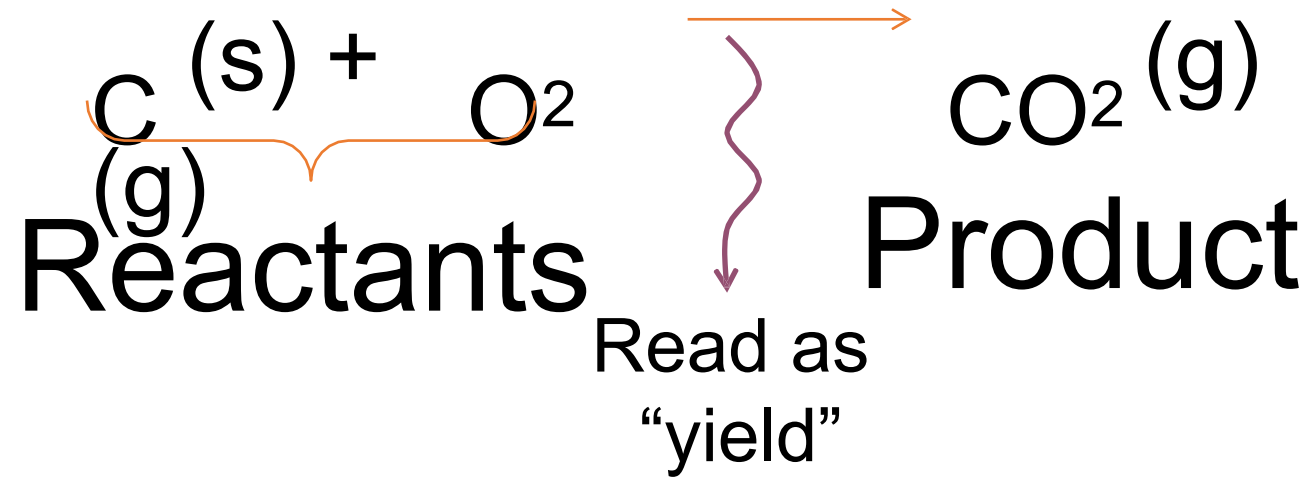


Chemical equation denotes a chemical reaction:



x , y , z , and w = stoichiometric coefficients

Example of Chemical Equation



At times we need to indicate the **physical state** of the **reactants and products** and the following symbols can be used.

(g) = gas (l) = liquid (aq) = aqueous (s) = solid

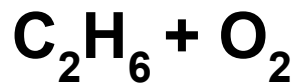
Why a balanced chemical equation is important?

It can be used to **calculate the correct amount of reactants required** and also to anticipate the **amount of products formed** by the reaction.

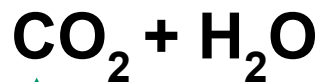
Chemical equation can be balanced by :

☐ Inspection Method

1. Write down the unbalanced equation. Write the **correct formulae** for the reactants and products.
2. Balance :
 - i. the **metallic element**, followed by **non-metallic atoms**.
 - ii. Balance the **hydrogen** and **oxygen** atoms.
3. Check to ensure that the total number of atoms of each element is the **same on both sides** of equation.

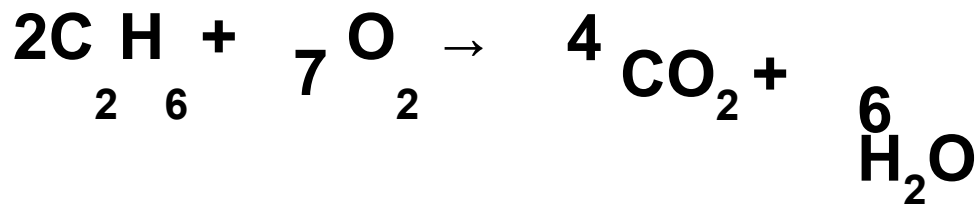
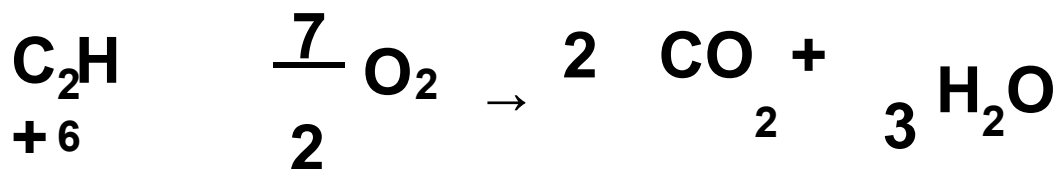
Example 1:**Inspection Method**

2 carbon
on the left



1 carbon
on the right

Start with C or H but not O



Reactants

Products

4 C

4 C

12 H

12 H

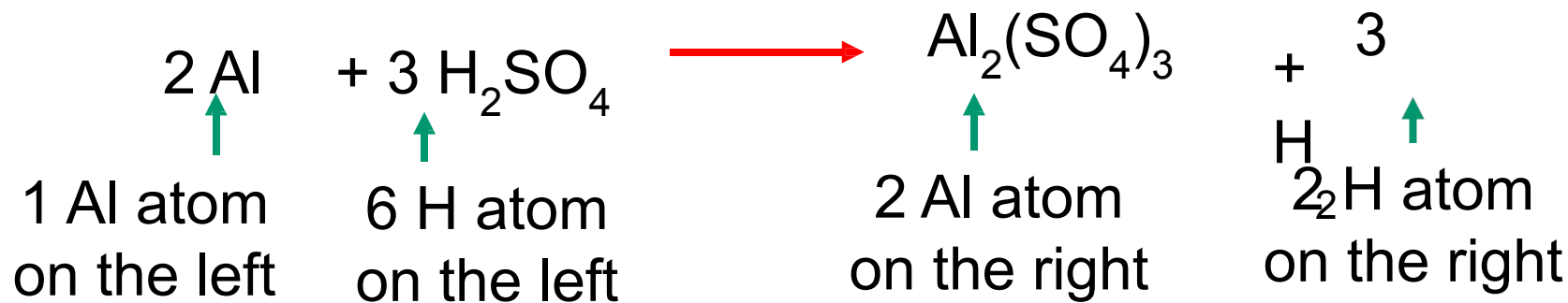
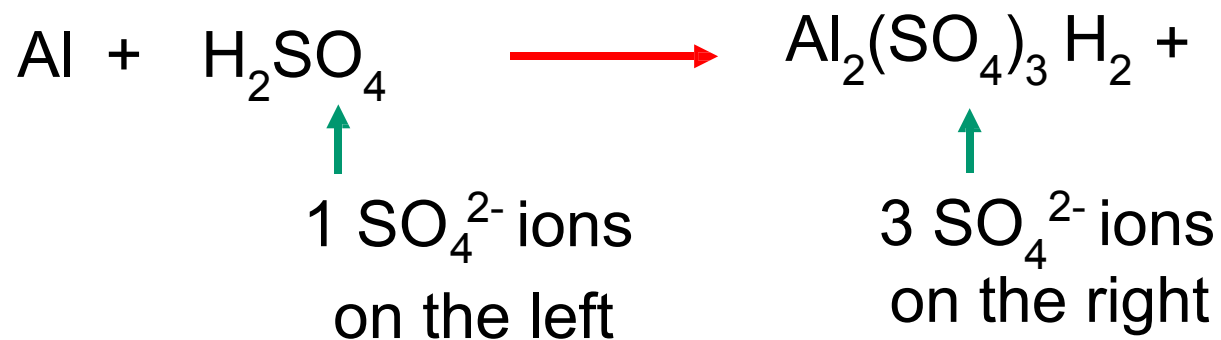
14 O

14 O

Example 2:

Balance polyatomic ions as a unit

Balance pure elements such as Al and H last.



Reactants

Products

2

2

A

Al

1

6 H

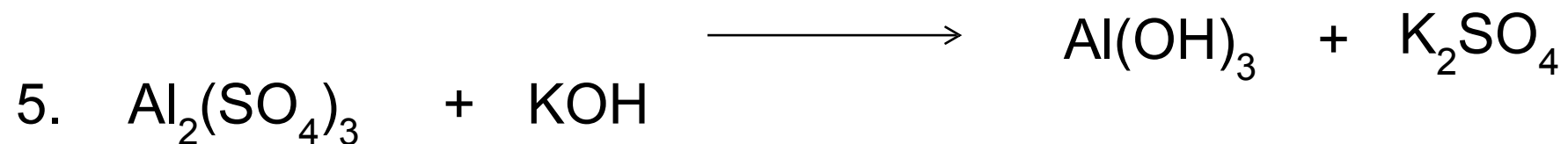
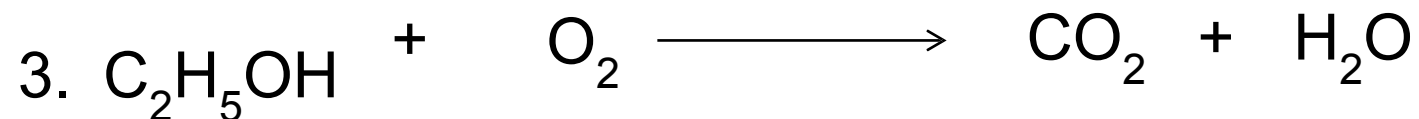
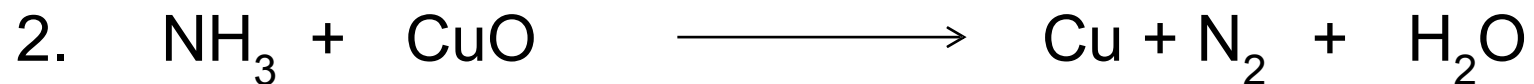
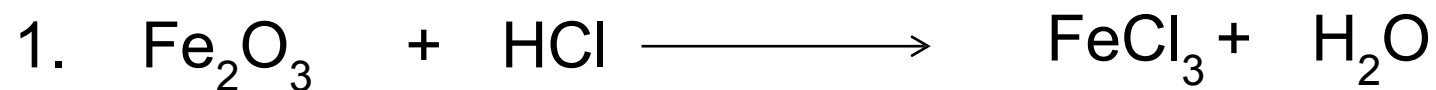
6H

12 S 12 O

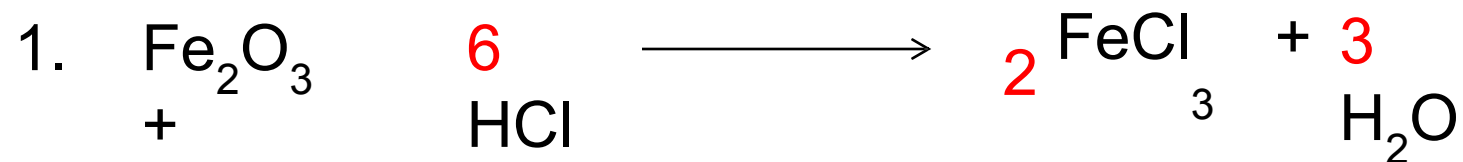
3 S

Practice Exercise

Balance the following chemical equations (by inspection method)



Answer:



Chemical equation can be balanced by :

❑ Ion-electron Method

- This method is used in balancing redox equation.
- Redox reaction is a reaction that involves both **reduction** and **oxidation** reactions.
- Redox reaction may occurs in **acidic** or **basic medium**.

Oxidation	Reduction
An increase in oxidation number	A decrease in oxidation number
Loss of electrons	Gain of electrons



OIL -Oxidation Is the Loss of electrons

RIG -Reduction Is the Gain of electrons

Oxidation Number

- Oxidation number is the charge the atom would have in a molecule (or an ionic compound) if electrons were completely transferred.

1. Oxidation number of the atom in any free elements is zero.

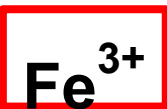
Example

Oxidation number of i. **Monatomic element**

K=0, Be=0, Al=0

ii. **Diatomic element**

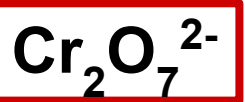
2. In monoatomic ions, oxidation number is equal to the charge on the ion.



Oxidation number of Fe = +3

$\text{O}_2 = 0$, $\text{Cl}_2 = 0$,
 $\text{F}_2 = 0$

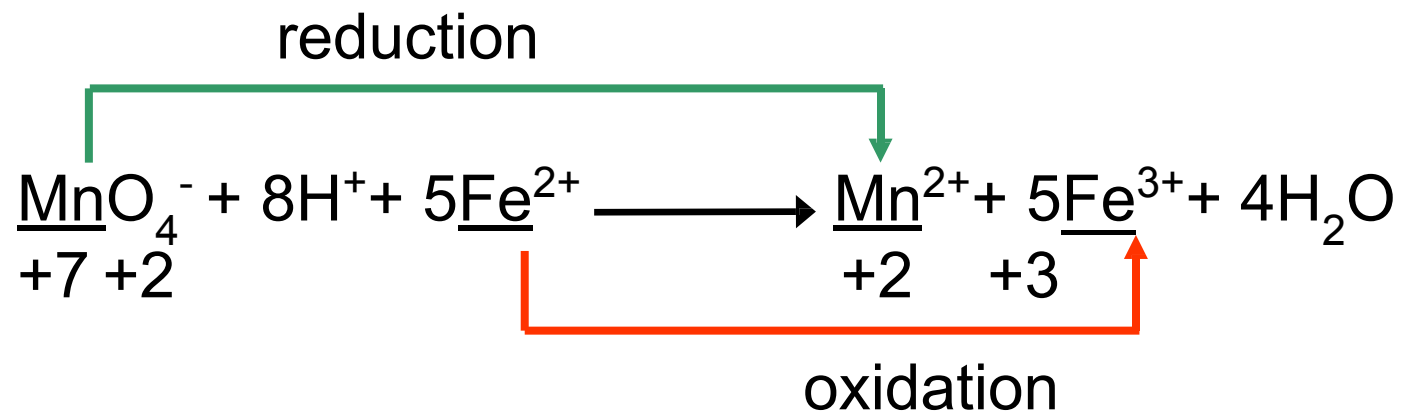
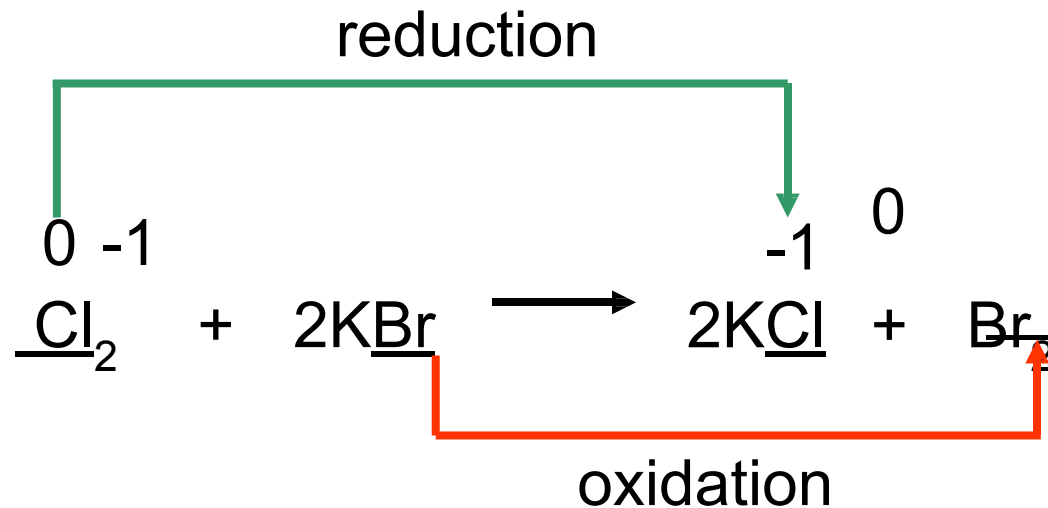
3. The oxidation number of other elements are then assigned to make the sum of the oxidation numbers equal to the net charge on the molecule or ion.



Total oxidation number = $2(+6) + 7(-2) = -2$

Example:

Redox equations :

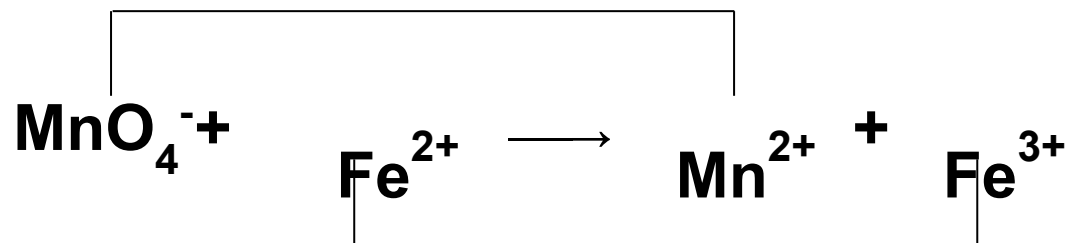
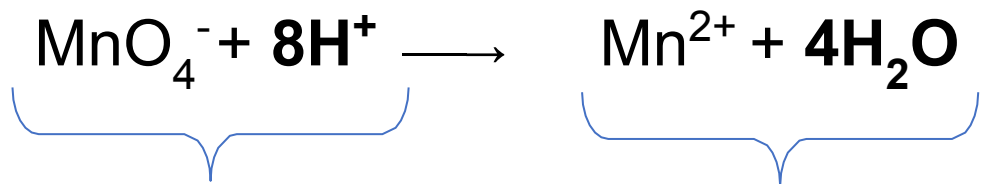
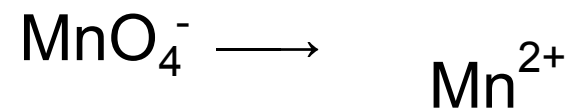


Balancing Redox Equations For Acidic Solutions

The sequence of steps in balancing redox equations :

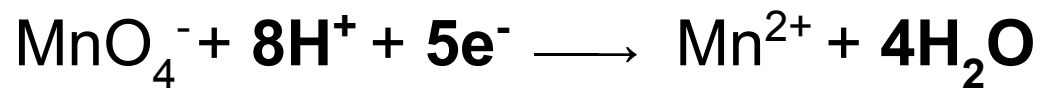
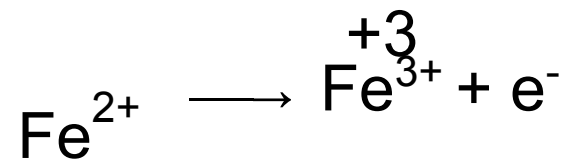
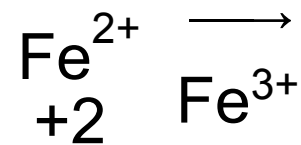
1. Divide the equation into **two half-equations**.
2. Balance the atoms other than H and O
3. Balance O by **adding H_2O** . Balance H by **adding H^+** .
4. Balance the net charge in each half-equation by **adding electron to the side that has more +ve charge**.
5. **Multiply the half-equations** by appropriate integers to make **number of $\bar{e}$ gain equals to the number of $\bar{e}$ loss**.
6. **Add two half-equations** in an overall equation. Cancel any species appearing on both sides.

*** Check whether the net charge and atoms are balanced on both sides of the equation.**

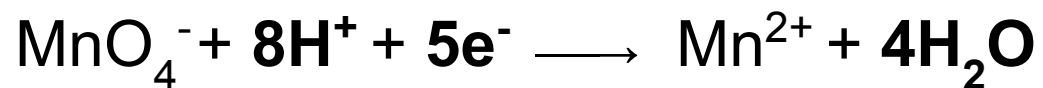
Example 1:**(Acidic solution)****Solution:****Reduction :**

$$-1 + 8 = +7$$

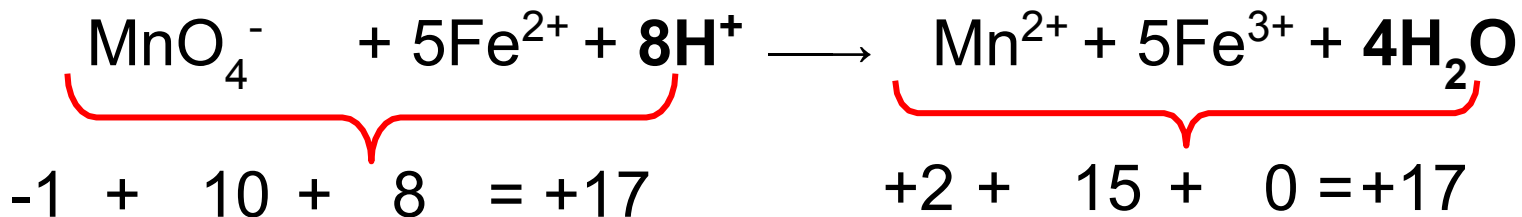
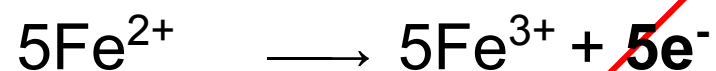
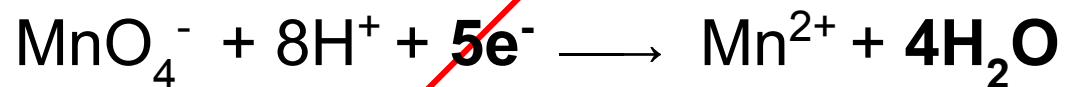
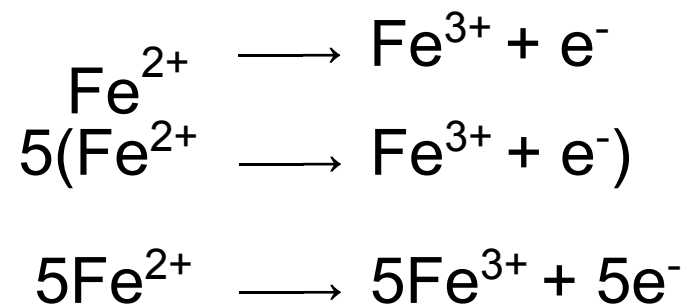
$$2 + 0 = +2$$

**Oxidation :**

Reduction :



Oxidation :

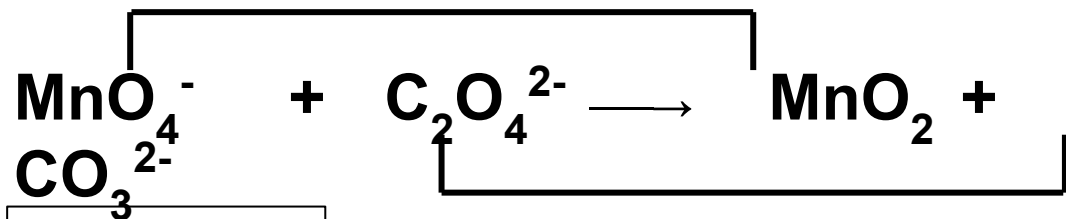


 ***Check for net charge and atoms***

Balancing Redox Equations For Basic Solutions

- **Assume the solution were acidic.** Balance the equation as if it is in an acidic solution.
- **Then follow a simple three-step :**
 1. Add to both sides of the equation the **same number of OH^-** as there are H^+ to eliminate ALL H^+
 2. **Combine H^+ and OH^-** to form H_2O .
 3. **Cancel any H_2O** that you can.

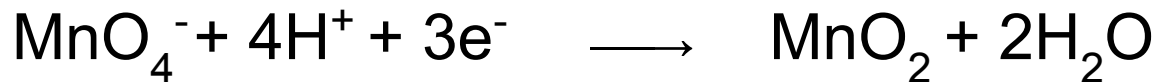
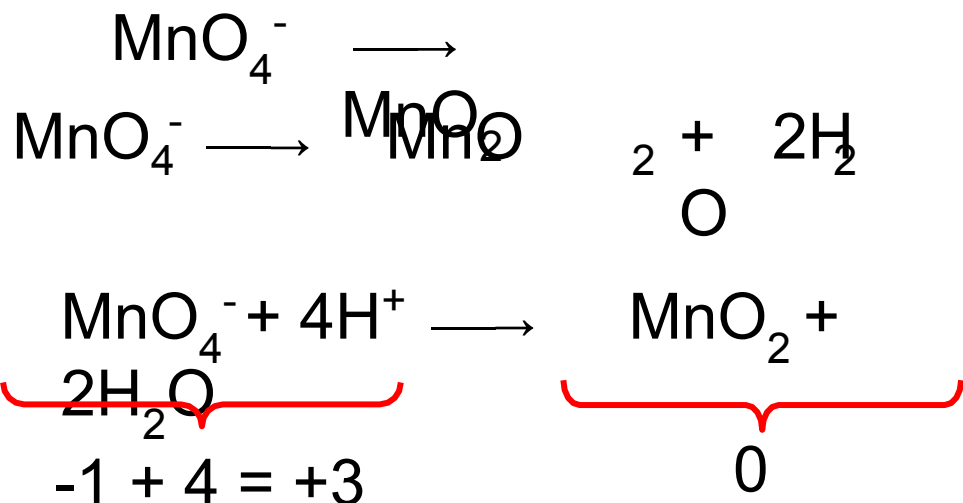
Example 2: (Basic solution)



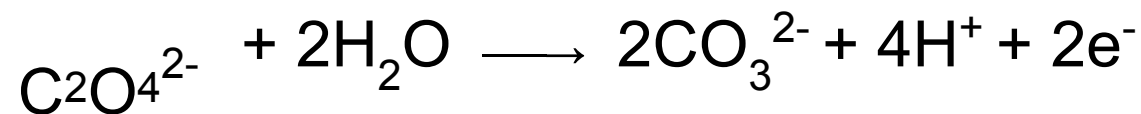
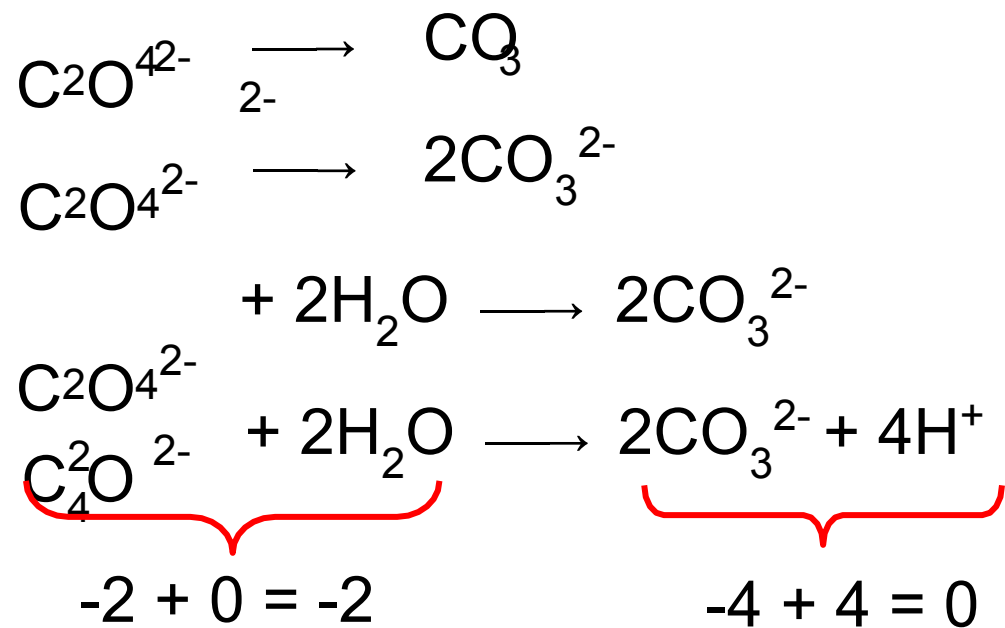
Solution:

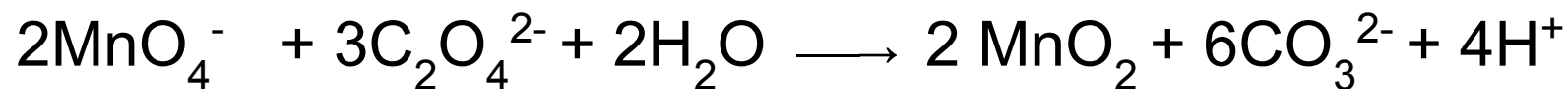
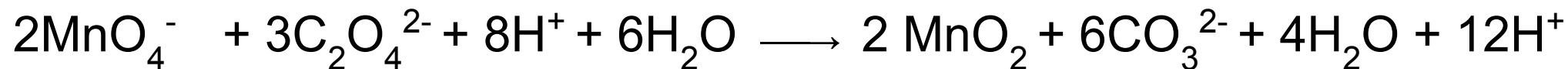
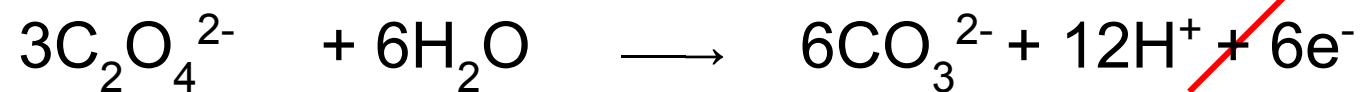
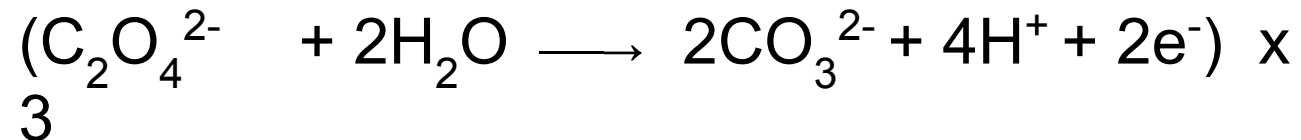
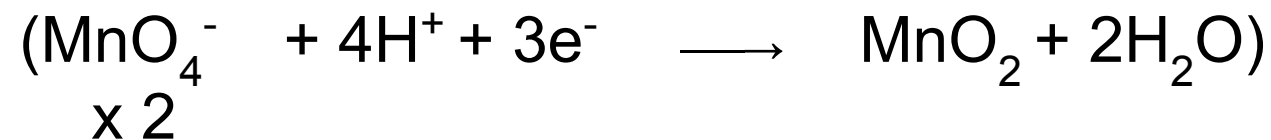
Assume in acidic solution

Reduction :



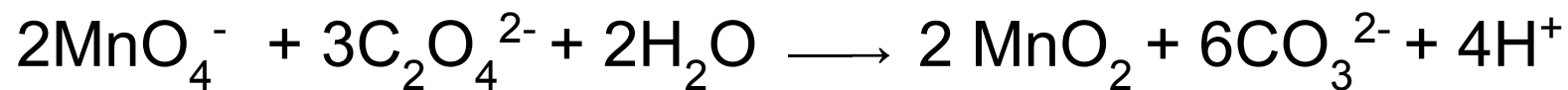
Oxidation :



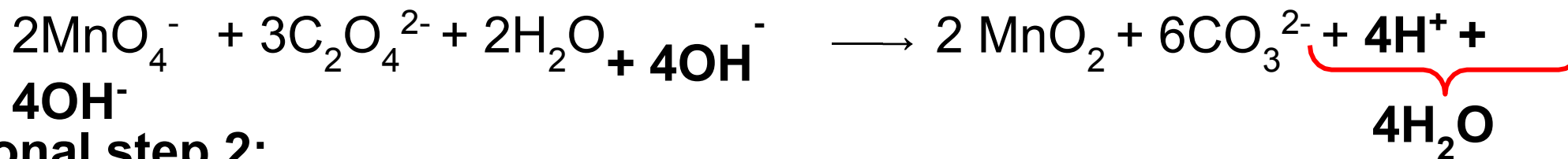


 Check for net charge and atoms

However, in basic solution



Additional step 1:



Additional step 2:

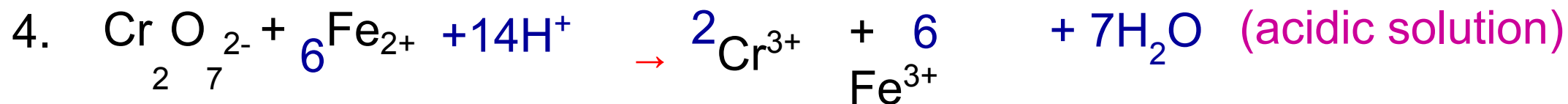
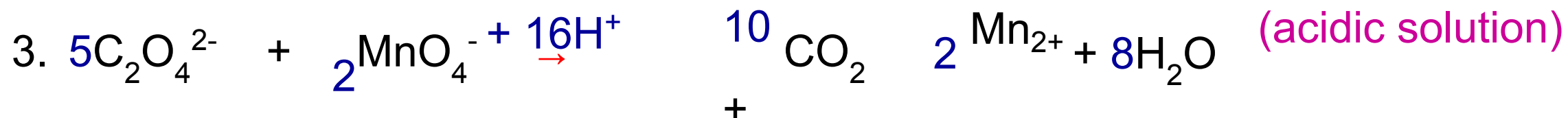
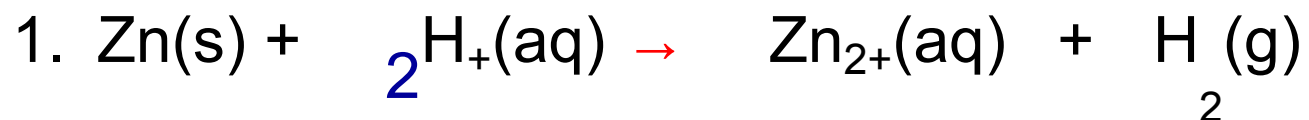


Additional step 3:



Practice Exercise

Balancing the following Redox Equations:

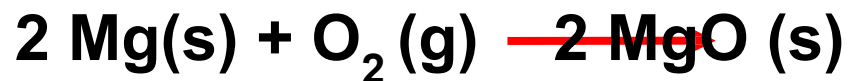


- Once a chemical equation is balanced, it is valuable to figure out all sorts of “how much” questions.
 - ☐ How much of each reactant must I mix together to make certain amounts of products?
 - ☐ How much products will I make if I mix certain amounts of reactants together?
 - ☐ Will any of the reactants be left over, and if so which one and how much?
- The study of the quantitative aspects of a chemical reactions are the subject of STOICHIOMETRY.

Stoichiometry : The quantitative study of reactants and products in a chemical reaction.

How to interpret chemical equations ?

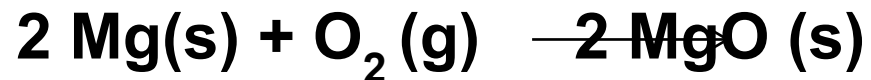
Example



- 2 Mg atoms + 1 O₂ molecule forms 2 formula units of MgO
- 2 mol Mg + 1 mol O₂ forms 2 mol MgO
- 48.6 g Mg + 32.0 g O₂ forms 80.6 g MgO

Stoichiometric Calculations

Example 1:



[$A_r \text{ O} = 16.0$, $\text{Mg} = 24.3$, 1 mol of gas occupies a volume of 22.4 dm^3 at STP]

- i. How many **moles of MgO** result from the complete reaction of **3.4 mol of O_2** ?
Assume there is more than enough Mg.

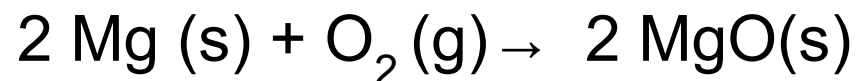
Solution:

From equation, 1 mol $\text{O}_2 \equiv 2 \text{ mol MgO}$

$$\begin{aligned} 3.4 \text{ mol O}_2 &\equiv \frac{3.4 \cancel{\text{ mol O}_2}}{1 \cancel{\text{ mol O}_2}} \times 2 \text{ mol MgO} \\ &= \mathbf{6.8 \text{ mol}} \end{aligned}$$

ii. How many grams of MgO formed when 20.0 g of Mg reacts completely with O₂ ?

Solution:



$$\text{Moles of Mg} = \frac{20.0 \text{ g}}{24.3 \text{ g mol}^{-1}} = 0.823$$

From equation:



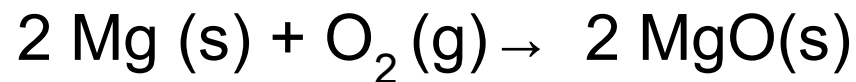
$$\therefore 0.823 \text{ mol Mg} \equiv \frac{0.823 \text{ mol Mg} \times 2 \text{ mol MgO}}{2 \text{ mol Mg}} = \mathbf{0.823 \text{ mol MgO}}$$

$$\text{Mass of MgO formed} = 0.823 \text{ mol} \times 40.3 \text{ g mol}^{-1}$$

$$= \mathbf{33.17 \text{ g}}$$

iii. How many litres of O₂ gas are used at STP when 40.0 g of MgO formed ?

Solution:



$$\text{Moles of MgO} = \frac{40.0 \text{ g}}{40.3 \text{ g mol}^{-1}} = 0.9926 \text{ mol}$$

From equation:

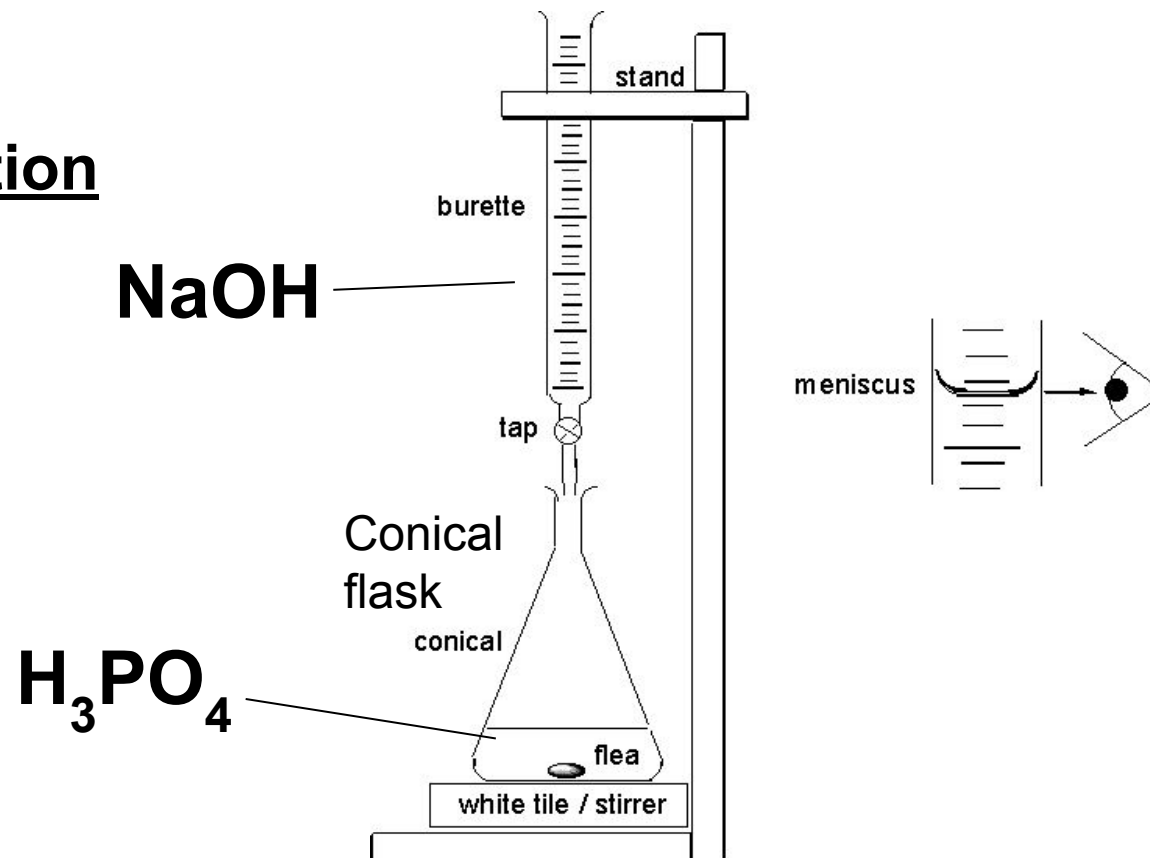
$$2 \text{ mol MgO} \equiv 1 \text{ mol of O}_2$$

$$\therefore 0.9926 \text{ mol MgO} \equiv \frac{0.9926 \cancel{\text{ mol MgO}}}{2 \cancel{\text{ mol MgO}}} \times 1 \text{ mol O}_2 = \mathbf{0.4963 \text{ mol O}_2}$$

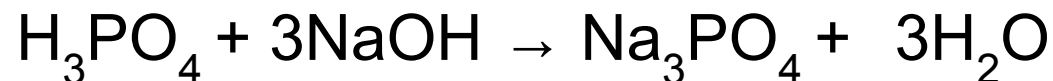
$$\text{Volume of O}_2 \text{ used} = \frac{\text{mol}_1}{\text{mol}_1} \times 22.4 \text{ L}$$

Example 2:

Acid-Base Titration



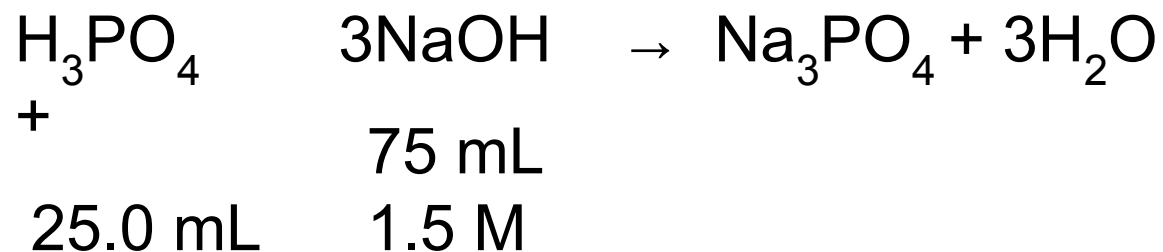
75 mL of a 1.5 M NaOH solution is required to neutralise 25.0 mL of a H₃PO₄ solution.



- Calculate the number of moles of NaOH used.
- Determine the molarity of the H₃PO₄ solution.

i. Calculate the number of moles of NaOH used.

Solution:

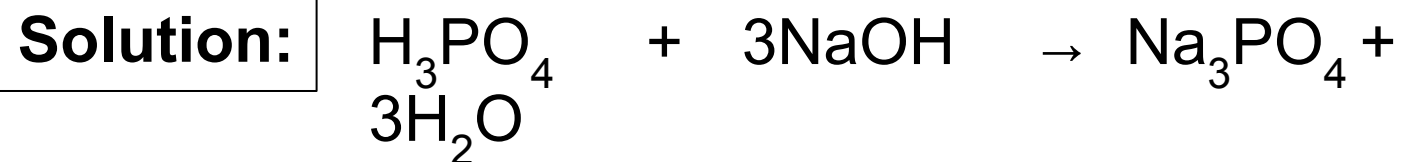


$$\text{Molarity} = \frac{\text{moles of solute}}{\text{volume of solution (L)}}$$

$$\begin{aligned} \therefore \text{moles of NaOH} &= MV \\ &= 1.5 \text{ mol L}^{-1} \times 0.075 \text{ L} \\ &= \\ &= \mathbf{0.1125} \\ &\mathbf{\text{mol}} \end{aligned}$$

ii. Determine the molarity of the H_3PO_4 solution.

Solution:



From equation, 3 mol NaOH $\equiv$ 1 mol H_3PO_4

$$0.1125 \text{ mol NaOH} \equiv \frac{0.1125 \text{ mol NaOH}}{3 \text{ mol NaOH}} \times 1 \text{ mol } \text{H}_3\text{PO}_4$$

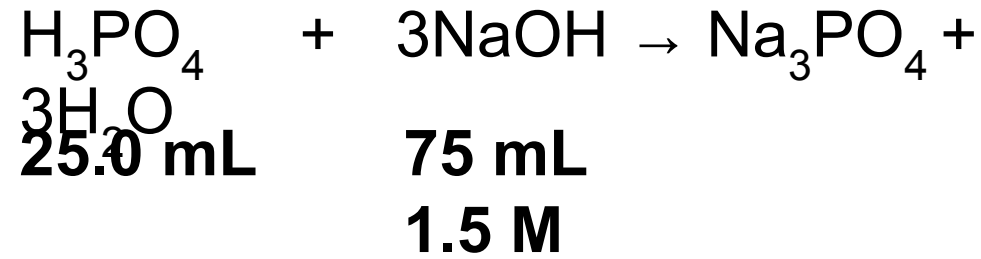
$$= 0.0375 \text{ mol } \text{H}_3\text{PO}_4$$

$$\begin{aligned} \text{Thus, molarity of } \text{H}_3\text{PO}_4 &= \frac{\text{moles of } \text{H}_3\text{PO}_4}{\text{volume of solution (L)}} \\ &= \frac{0.0375 \text{ mol}}{25 \times 10^{-3} \text{ L}} \end{aligned}$$

Alternative method



$$\frac{M_A V_A}{M_B V_B} = \frac{a}{b}$$



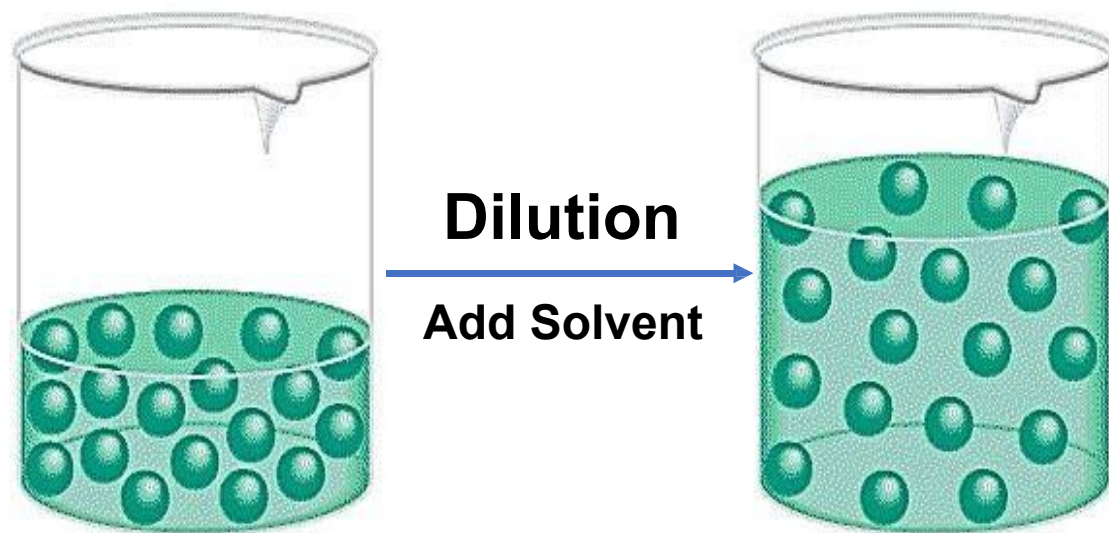
$$\Rightarrow \frac{M_{\text{H}_3\text{PO}_4} V_{\text{H}_3\text{PO}_4} \frac{1}{3}}{M_{\text{NaOH}} V_{\text{NaOH}}} =$$

$$\frac{M_{\text{H}_3\text{PO}_4} (0.025\text{L})}{(1.5\text{M})(0.075\text{L})} \frac{1}{3} =$$

$$M_{\text{H}_3\text{PO}_4} = \frac{(1.5\text{M})(0.075\text{L})}{3(0.025\text{L})} = 1.5 \text{ M}$$

Dilution of Solution

Dilution is the procedure for preparing a less concentrated solution from a more concentrated solution.



**Moles of solute
before dilution**

=

**Moles of solute
after dilution**

$$M_i V_i$$

=

$$M_f V_f$$

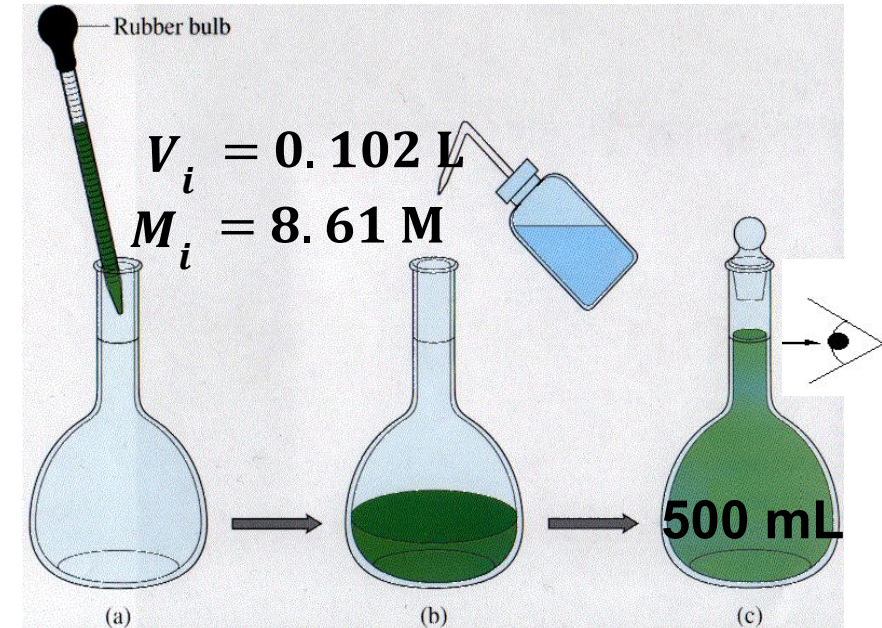
Example 3:

Describe how would you prepare 0.50 L of 1.75 M HCl solution, starting with an 8.61 M stock solution of HCl.

Solution:

$$\begin{aligned}M_i &= 8.61 \text{ M} \\V_i &= ? \\M_f &= 1.75 \text{ M} \\V_f &= 0.5 \text{ L}\end{aligned}$$

$$\begin{aligned}M_i V_i &= M_f V_f \\(8.61 \text{ M}) V_i &= (1.75 \text{ M})(0.50 \text{ L}) \\V_i &= \mathbf{0.102 \text{ L}}\end{aligned}$$



We must dilute 0.102 L of 8.61 M HCl solution with sufficient water to give a final volume of 0.50 L in 500 mL volumetric flask to obtain the desired concentration.

$$\therefore \text{Volume of water added} = 0.50 - 0.102 = \mathbf{0.398 \text{ L @ } 398 \text{ mL}}$$

Example 4: PSPM 2009/2010

An amount of 27.82 g of hydrated sodium carbonate crystals, $\text{Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O}$, was dissolved in water and made up to 1 L. 25.00 mL of this solution was neutralised by 48.80 mL of 0.100 M hydrochloric acid.

- a) Write an equation for the reaction between sodium carbonate and hydrochloric acid.

[1 mark]

- b) Find the molar concentration of sodium carbonate neutralised by the hydrochloric acid and find the value of x .

[7 marks]

- c) If 5.00×10^{-2} M sulphuric acid were used instead of hydrochloric acid in the above analysis, what would be the volume of the sulphuric acid needed?

[2 marks]

a) Write an equation for the reaction between sodium carbonate and hydrochloric acid.

[1 mark]

Solution:



b) Find the molar concentration of sodium carbonate neutralised by the hydrochloric acid and find the value of x.

[7 marks]

Solution:

Method 1

$$\begin{aligned}\text{Mole of HCl} &= MV \\ &= 48.80 \times 10^{-3} \text{ L} \times 0.1 \text{ mol L}^{-1} \\ &= 4.88 \times 10^{-3} \text{ mol}\end{aligned}$$

From equation, $2 \text{ mol HCl} \equiv 1 \text{ mol Na}_2\text{CO}_3$

$$\begin{aligned}4.88 \times 10^{-3} \text{ mol HCl} &= \frac{4.88 \times 10^{-3} \text{ mol HCl} \times 1 \text{ mol Na}_2\text{CO}_3}{2 \text{ mol HCl}} = 2.44 \times 10^{-3} \text{ mol Na}_2\text{CO}_3\end{aligned}$$

Continue

$$1000 \text{ mL} = 1 \text{ L}$$

$$25.00 \text{ mL} = 25 \times 10^{-3} \text{ L}$$

$$\begin{aligned} \text{Concentration of Na}_2\text{CO}_3 &= \frac{\text{mole Na}_2\text{CO}_3}{\text{L of solution}} \\ &= \frac{2.44 \times 10^{-3}}{25 \times 10^{-3} \text{ L}} \\ &= 9.76 \times 10^{-2} \text{ M} \end{aligned}$$

Method 2

$$M_{\text{Na}_2\text{CO}_3} V_{\text{Na}_2\text{CO}_3} = \frac{1}{2} M_{\text{HCl}} V_{\text{HCl}}$$

$$\frac{M_{\text{Na}_2\text{CO}_3} (0.025)}{(0.1)(0.0488)} = \frac{1}{2}$$

$$M_{\text{Na}_2\text{CO}_3} = 9.76 \times 10^{-2} \text{ M}$$

Mol of $\text{Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O}$ in 1 L solution
=MV

$$=9.76 \times 10^{-2} \text{ mol L}^{-1} \times 1 \text{ L}$$

$$=9.76 \times 10^{-2} \text{ mol}$$

$$27.82 \text{ g Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O} \equiv 9.76 \times 10^{-2} \text{ mol}$$

$$?? \text{ g Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O} \equiv 1 \text{ mol}$$

$$\text{Molar mass of Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O} = \frac{27.82 \text{ g}}{9.76 \times 10^{-2} \text{ mol}}$$

$$=285 \text{ g mol}^{-1}$$

$$\text{Molar mass of Na}_2\text{CO}_3 \cdot x\text{H}_2\text{O} = 2(23) + 12 + 3(16) + x(18)$$

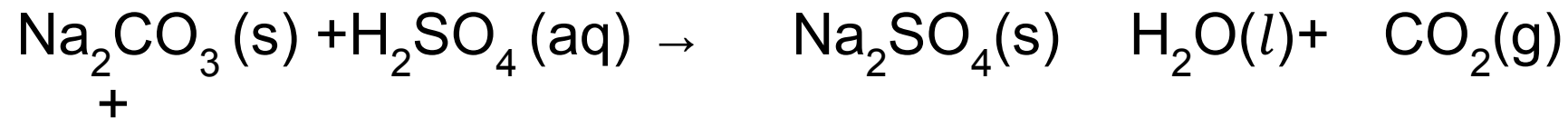
$$285 = 106 + 18x$$

$$x = 10$$

c) If $5.00 \times 10^{-2} \text{ M}$ sulphuric acid were used instead of hydrochloric acid in the above analysis, what would be the volume of the sulphuric acid needed?

[2 marks]

Solution:



$$\frac{M_a V_a}{M_b V_b} = \frac{a}{b} \quad a = \text{Na}_2\text{CO}_3; b = \text{H}_2\text{SO}_4$$

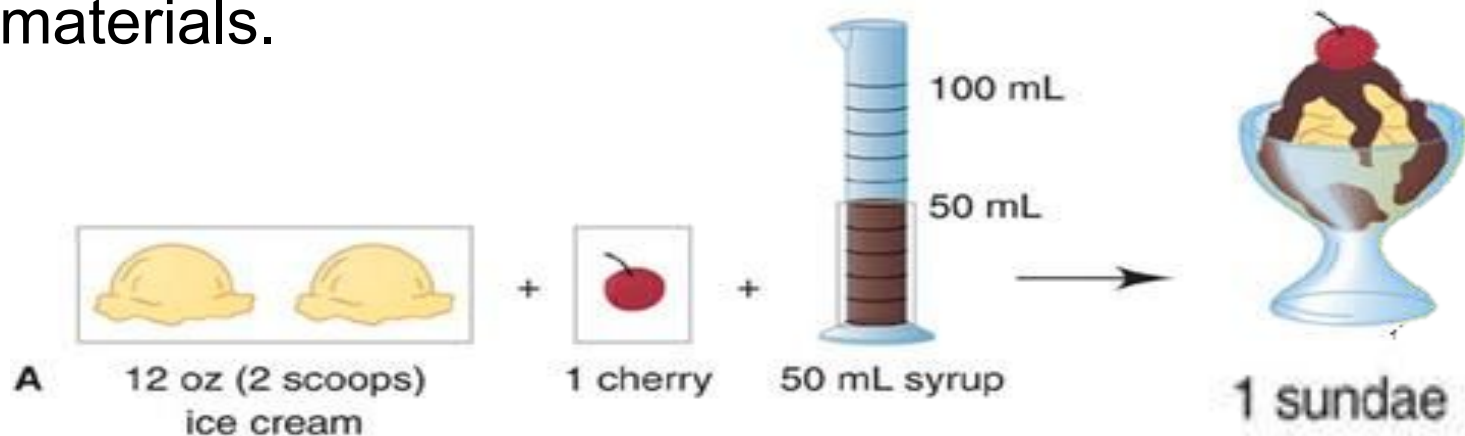
$$\frac{M_{\text{Na}_2\text{CO}_3} V_{\text{Na}_2\text{CO}_3}}{M_{\text{H}_2\text{SO}_4} V_{\text{H}_2\text{SO}_4}} = \frac{1}{1}$$

$$\frac{(9.76 \times 10^{-2} \text{ M})(0.025 \text{ L})}{(5.00 \times 10^{-2} \text{ M}) V_{\text{H}_2\text{SO}_4}} = \frac{1}{1}$$

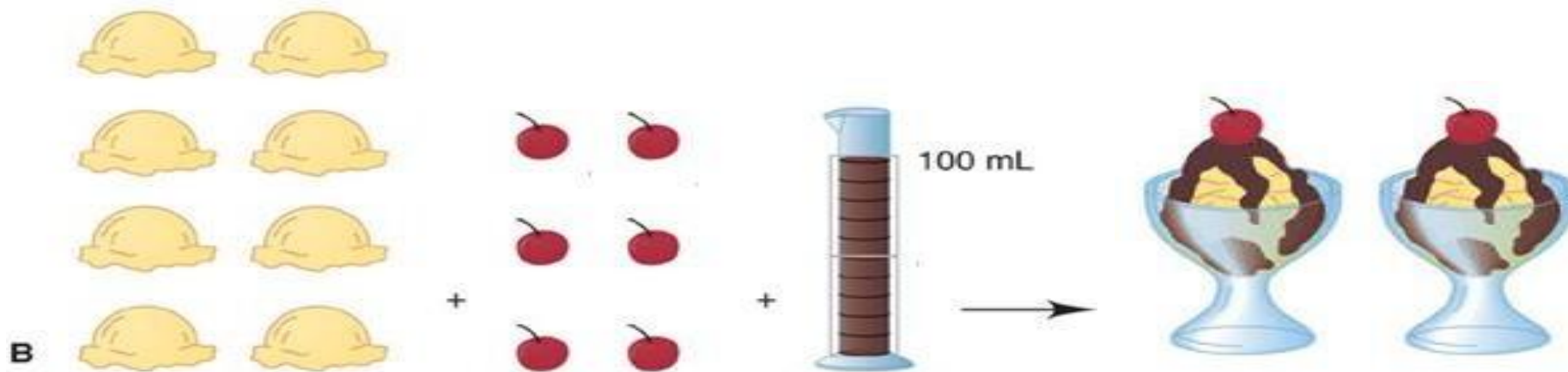
$$V_{\text{H}_2\text{SO}_4} = \mathbf{0.0488 \text{ L}}$$

Limiting Reactant & Percentage Yield

A cup of sundae can be made by using the following proportions of materials.



Therefore, only 2 cups of sundae can be made if the following materials are provided.



Which one limits the number of cups of sundae ?

A
N
A
L
O
G
Y

If this were a chemical reaction,
the syrup would be the limiting reactant or limiting reagent.

A **limiting reactant** is the *reactant that is completely consumed* in a chemical reaction and *limits the amount of products formed*.

The reactants that do not limit the amount product, such as ice cream and cherry are in excess.

An **excess reactant** is the reactant that is *present in a quantity greater than is required* to completely react with the limiting reactant.



2 cups of sundae would be the **theoretical yield**.

Theoretical yield is the **amount of product that can be formed** in a chemical reaction based on the amount of limiting reactant.

Suppose we accidentally spill the syrup and can only come up with 1 cup of sundae



1 cup of sundae would be the **actual yield**.

Actual yield is the **amount of product actually obtained** from a reaction.

The actual yield is always *equal to or less than* the theoretical yield because a small amount of product is usually lost to other reaction or does not form during a reaction.

Percentage yield can be calculated as follows :

$$\text{Percentage yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100\%$$

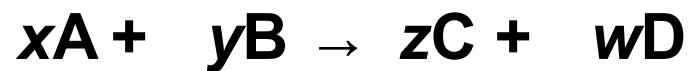
obtained

calculated

In the making of sundae,

$$\text{percentage yield} = \frac{1 \text{ sundae}}{2 \text{ sundae}} \times 100\% = 50\%$$

we obtained only 50% of our theoretical yield.

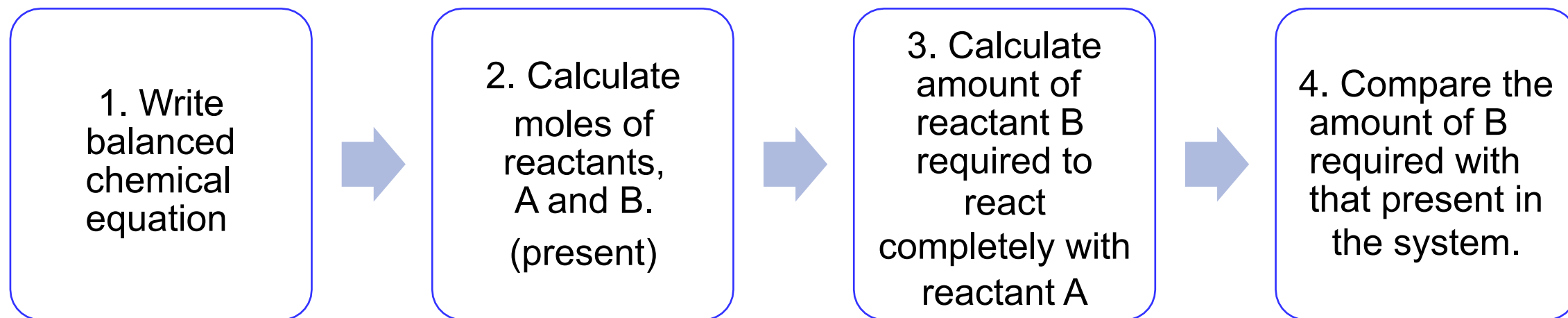


reactants

products

Determining Limiting Reactant

Summary of Stoichiometric Method



If mol of B (present) < mol of B(required)

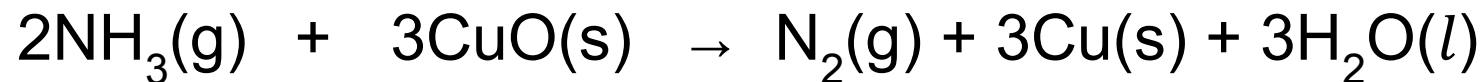
$\therefore$ **B = limiting reactant**

If mol of B (present) > mol of B(required)

$\therefore$ **B = excess reactant $\Rightarrow$ A = limiting reactant**

Example 1:

For the following reaction :



18.0 g NH_3 and 90.0 g CuO are allowed to react.

[$A_r \text{ Cu} = 63.6$, $\text{H} = 1.0$, $\text{N} = 14.0$, $\text{O} = 16.0$]

- (a) Determine the limiting reactant.
- (b) Calculate the mass of N_2 gas formed.
- (c) Determine the mass of the excess reactant remain after the completion of the reaction.

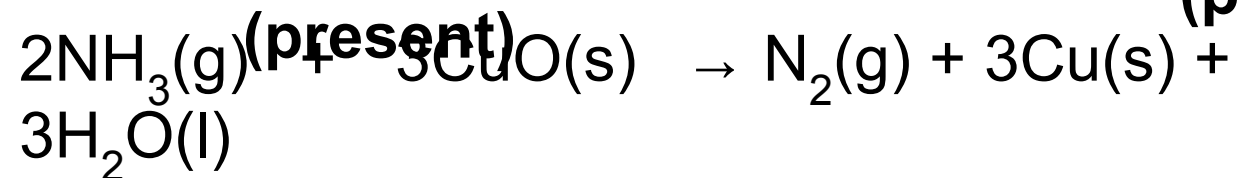
Solution:

(a) Determine the limiting reactant.

$$n_{\text{NH}_3} = \frac{18 \text{ g}}{17 \text{ g mol}^{-1}} \\ = 1.059 \text{ mol}$$

$$n_{\text{CuO}} = \frac{90 \text{ g}}{79.6 \text{ g mol}^{-1}} \\ = 1.131 \text{ mol}$$

(present)



From equation,

$$2 \text{ mol NH}_3 \equiv 3 \text{ mol CuO}$$

$$\therefore 1.059 \text{ mol NH}_3 \equiv \frac{1.059 \text{ mol NH}_3 \times 3 \text{ mol CuO}}{2 \text{ mol NH}_3}$$

$$n_{\text{CuO}} (\text{present}) = 1.131 \text{ mol} \\ = 1.589 \text{ mol CuO (required)}$$
$$n_{\text{CuO}} (\text{present}) < n_{\text{CuO}} (\text{required})$$

$\therefore$ CuO is the limiting reactant.

Solution: (b) Calculate the mass of N_2 gas formed.

From equation,



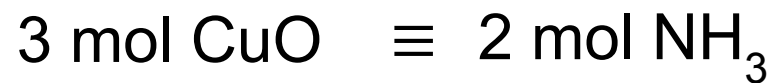
$$\begin{aligned} 1.131 \text{ mol CuO} &\equiv \frac{1.131 \text{ mol CuO}}{3 \text{ mol CuO}} \times 1 \text{ mol N}_2 \\ &\equiv 0.377 \text{ mol N}_2 \end{aligned}$$

$$\begin{aligned} \text{Mass of N}_2 \text{ formed} &= 0.377 \text{ mol} \times 28 \text{ g mol}^{-1} \\ &= \mathbf{10.556 \text{ g}} \end{aligned}$$

Solution:

(c) Determine the mass of the excess reactant remain after the completion of the reaction.

From equation,



$$1.131 \text{ mol CuO} \equiv \frac{1.131 \text{ mol CuO}}{3 \text{ mol CuO}} \times 2 \text{ mol NH}_3$$

$$\equiv 0.754 \text{ mol NH}_3 \text{ (reacted)}$$

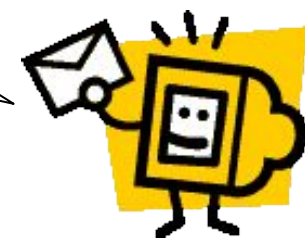
$$\text{Mass of NH}_3 \text{ reacted} = 0.754 \text{ mol} \times 17 \text{ g mol}^{-1}$$

$$= 12.818 \text{ g}$$

$$\therefore \text{Mass of NH}_3 \text{ in excess} = 18 - 12.818$$

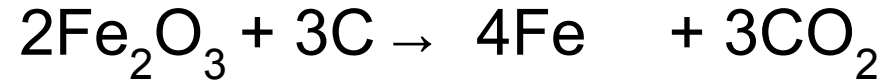
$$= 5.182 \text{ g}$$

Only the limiting reactant is completely consumed



Example 2:

The reaction of iron ore with carbon follows the equation:

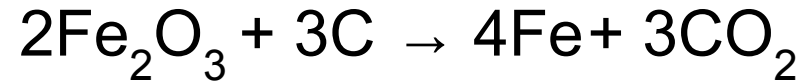


200 g Fe_2O_3 and 300 g C react to produce 110 g Fe.

- (a) Calculate the theoretical yield of Fe.
- (b) Calculate the percent yield of Fe.

Solution:

(a) Calculate the theoretical yield of Fe.



$$n_{\text{Fe}_2\text{O}_3} = \frac{200 \text{ g}}{159.8 \text{ g mol}^{-1}} = 1.252 \text{ mol}$$

(present)

$$n_{\text{C}} = \frac{300 \text{ g}}{12 \text{ g mol}^{-1}} = 25 \text{ mol}$$

(present)

cont:

Determine the limiting reactant

From equation,

$$\begin{aligned} 2 \text{ mol Fe}_2\text{O}_3 &\equiv 3 \text{ mol C} \\ 1.252 \text{ mol Fe}_2\text{O}_3 &\equiv \frac{1.252 \text{ mol}}{2 \text{ mol Fe}_2\text{O}_3} \times 3 \text{ mol C} \\ &= 1.878 \text{ mol C (required)} \end{aligned}$$

$n_{\text{C (present)}} > n_{\text{C (required)}}$; C is excess reactant.
 $\therefore \text{Fe}_2\text{O}_3$ is the limiting reactant

Continue:

From equation,



$$1.252 \text{ mol Fe}_2\text{O}_3 \equiv \frac{1.252 \text{ mol Fe}_2\text{O}_3}{2 \text{ mol Fe}_2\text{O}_3} \times 4 \text{ mol Fe}$$
$$= 2.504 \text{ mol Fe}$$

$\therefore$ Mass of Fe formed theoretically,

$$= 2.504 \text{ mol} \times 55.9 \text{ g mol}^{-1}$$

$$= \mathbf{139.974 \text{ g}}$$

$$= \mathbf{140 \text{ g}}$$

Fe₂O₃ limits the amount of Fe formed

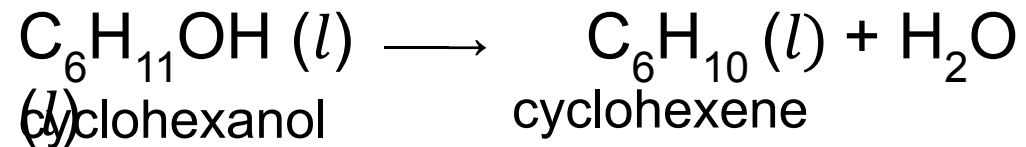


Solution: (b) Calculate the percent yield of Fe.

$$\text{Percentage yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100\%$$

$$= \frac{110 \text{ g}}{140 \text{ g}} \times 100\%$$

$$= \mathbf{78.57 \%}$$

Example 3:

If the percent yield of this reaction is 83 %, what mass of cyclohexanol must be used to obtain 25 g cyclohexene?

Solution:

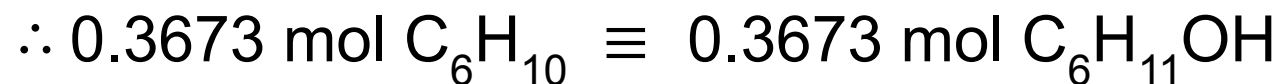
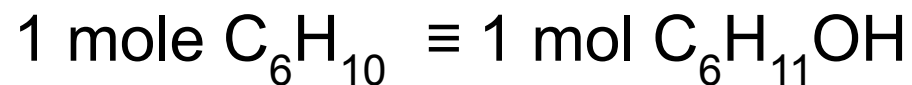
$$\begin{aligned} \text{Percentage yield of cyclohexene} &= \frac{\text{actual yield}}{\text{theoretical yield}} \times 100 \% \\ 83\% &= \frac{25}{\text{theoretical yield}} \times 100\% \end{aligned}$$

∴ Theoretical yield of cyclohexene = 30.12 g

$$\begin{aligned} \text{Moles of cyclohexene, } \text{C}_6\text{H}_{10} &= \frac{30.12 \text{ g}}{82.0 \text{ g}} \\ \text{mol}^{-1} &= 0.3673 \end{aligned}$$

Continue:

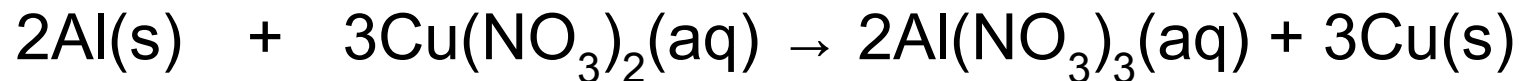
From equation, $\text{C}_6\text{H}_{11}\text{OH} (l) \rightarrow \text{C}_6\text{H}_{10} (l) + \text{H}_2\text{O} (l)$



$$\begin{aligned} \therefore \text{Mass of } \text{C}_6\text{H}_{11}\text{OH needed} &= 0.3673 \text{ mol} \times 100 \text{ g mol}^{-1} \\ &= \mathbf{36.73 \text{ g}} \end{aligned}$$

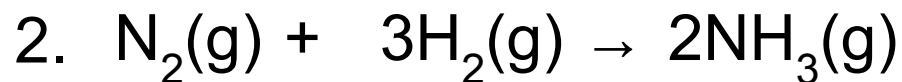
Practice Exercise

1. 0.500 g Al is allowed to react with 100 cm³ of 0.35 M Cu(NO₃)₂ solution.



- (a) Determine the limiting reactant.
- (b) Calculate the mass of Cu formed in the reaction.
- (c) Calculate the mass of the excess reactant after the reaction is completed.

(1.764 g, 1.36 g)



224.0 g of N₂ gas and 70.0 g of H₂ gas are mixed at temperature 400 °C and pressure 450 atm. After the completion of the reaction, 200 g of NH₃ is

Determine the percentage yield.

(73.53 %)

Practice Exercise

PSPM June 1999

3. Metallic aluminium, Al reacts with sulphuric acid, H_2SO_4 to produce aluminium sulphate, $\text{Al}_2(\text{SO}_4)_3$ and hydrogen gas, H_2 . If 10.0 g of aluminium is reacted with 60.0 g sulphuric acid to produce 57.0 g solid $\text{Al}_2(\text{SO}_4)_3$,

(a) Determine the limiting reactant.

(b) Calculate the theoretical yield of

(63.39 g)

(c) $\text{Al}_2(\text{SO}_4)_3$. Determine the percentage yield.

(89.92 %)

Practice Exercise PSPM June 2000

4. Hydrogen gas is produced when a certain mass of zinc reacts with hydrochloric acid. Hydrochloric acid provided by a supplier has a concentration of 12.5 M.
- a) Write a balance equation for the reaction involved in preparing hydrogen gas.
 - b) Calculate the volume of the concentrated hydrochloric acid needed to prepare 100 mL of 0.5 M hydrochloric acid. (4 mL)
 - c) If 2.356 g of zinc were reacted with 25 mL of 0.5 M hydrochloric acid, calculate the
 - (i) mass of hydrogen gas produced (0.0125 g)
 - (ii) volume of hydrogen gas produced at room temperature (0.15 L)

An abstract graphic on the left side of the image, consisting of overlapping, wavy, translucent shapes in shades of green, yellow, orange, red, and purple, resembling a stylized flower or a series of waves.

THE END